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Experiment #10 Amplitude Shift Keying (ASK).

Report #3

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Abstract

The aim of this experiment is to study Amplitude Shift Keying (ASK) in digital communication systems. The experiment explores two keys ASK methods, Hard Keying and Soft Keying, to understand their operating principles and differences. The effect of the modulation index on ASK performance is investigated in both the time and frequency domains, focusing on its impact on signal quality, noise susceptibility, and overall system efficiency. In addition, ASK modulation and demodulation are examined, and the effects of varying message amplitude, message frequency, and duty cycle are analysed. Through these investigations, the experiment aims to provide insights into optimizing ASK systems for practical digital communication applications.

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Theory

Amplitude shift keying (ASK)

Amplitude Shift Keying (ASK) is a digital modulation technique in which the amplitude of the carrier depends directly on the transmitted bit. When the input data bit is binary 1, the carrier is transmitted with a high (non-zero) amplitude. When the input data bit is binary 0, the carrier amplitude is low or completely absent (in the case of on–off Keying, OOK). [1]

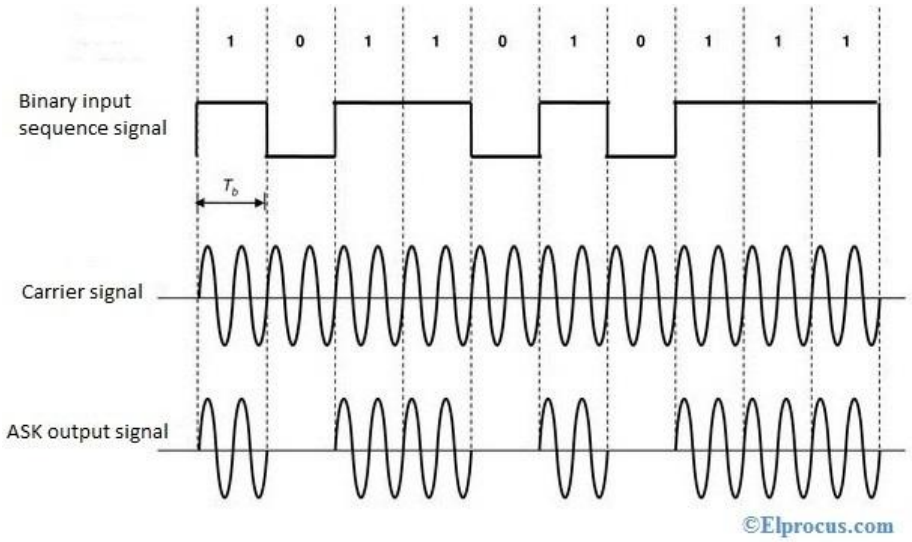


Figure 1: Amplitude shift keying.[2]

Soft and Hard Keying

The envelope of an ASK-modulated signal is determined by the shape of the baseband signal. When the baseband signal consists of rectangular impulses, the envelope changes abruptly, resulting in hard keying where the signal switches suddenly between two amplitude levels. In contrast, when the baseband signal is shaped using a Nyquist filter, the envelope transitions become smooth, a behaviour known as soft keying. These smooth envelope transitions reduce distortion and suppress unwanted high-frequency components, producing a cleaner spectrum and more efficient bandwidth usage. [3]

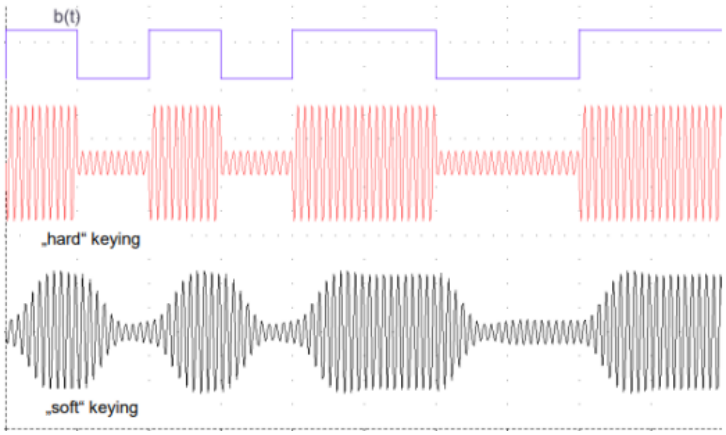


Figure 2: Soft and Hard Keying.[3]

Amplitude shift keying modulation

Amplitude Shift Keying (ASK) modulation is a digital modulation technique in which the amplitude of a high-frequency carrier signal is varied according to the binary input data. When the input data is binary 1, the carrier is transmitted with a constant non-zero amplitude. When the input data is binary 0, the carrier amplitude is reduced to zero or to a lower level, effectively turning the carrier off in the case of on–off keying (OOK). During ASK modulation, the frequency and phase of the carrier remain unchanged, and only the amplitude varies to represent the digital information.[4]

The ASK modulator block diagram comprises of the carrier signal generator, the binary sequence from the message signal and the band-limited filter. Following is the block diagram of the ASK Modulator.

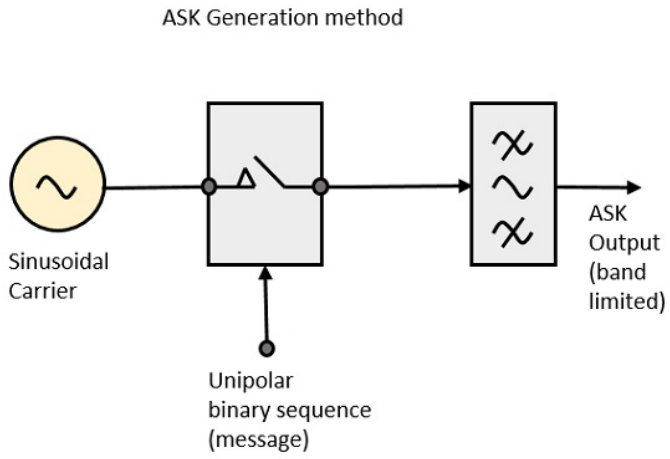


Figure 3: ASK Modulation Circuit.[4]

The carrier generator sends a continuous high-frequency carrier. The binary sequence from the message signal makes the unipolar input to be either High or Low. The high signal closes the switch, allowing a carrier wave. Hence, the output will be the carrier signal at high input. When there is low input, the switch opens, allowing no voltage to appear. Hence, the output will be low. The band-limiting filter shapes the pulse depending upon the amplitude and phase characteristics of the band-limiting filter or the pulse-shaping filter.[4]

Amplitude shift keying demodulation

- **Asynchronous (non – coherent) demodulation**

The asynchronous ASK detector consists of a half-wave rectifier, a low-pass filter, and a comparator. The received modulated ASK signal is first applied to the half-wave rectifier, which removes the negative portion of the carrier and produces a signal corresponding to the positive envelope. This rectified signal is then passed through a low-pass filter that suppresses the high-frequency carrier components and extracts the envelope of the ASK signal. Finally, the comparator compares the recovered envelope with a predefined threshold level and converts it into a digital output, representing logic 1s and 0s. [5]

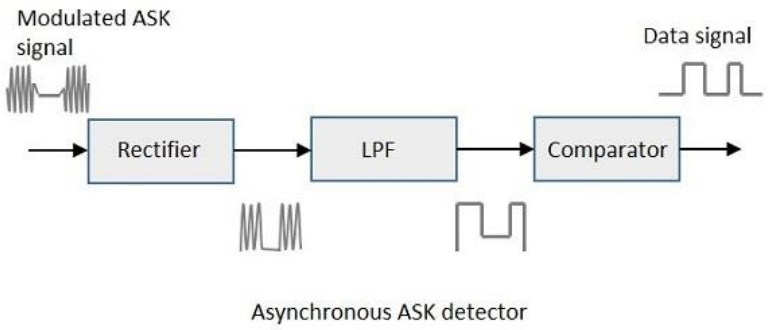


Figure 4: ASK Asynchronous Demodulation.[5]

- **Synchronous (coherent) demodulation**

The synchronous ASK demodulator consists of a square law detector, a low-pass filter, a comparator, and a voltage limiter. The received ASK-modulated signal is first applied to the square law detector, whose output voltage is proportional to the square of the amplitude of

the input signal. This operation converts the amplitude variations of the ASK signal into a baseband component. The low-pass filter then suppresses the high-frequency components and retains the desired baseband signal. Finally, the comparator and voltage limiter process this signal to remove unwanted fluctuations and produce a clean digital output corresponding to the transmitted binary data.[5]

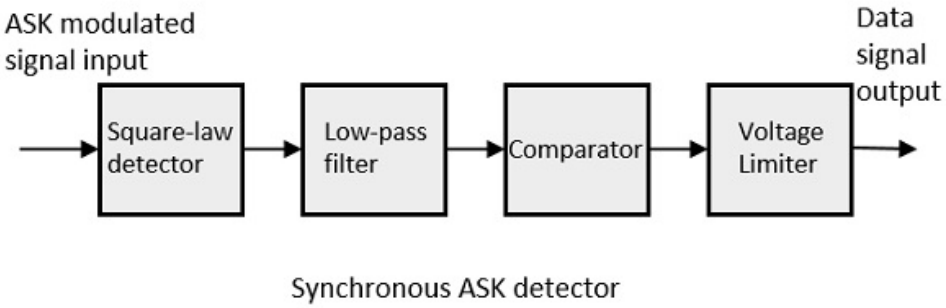
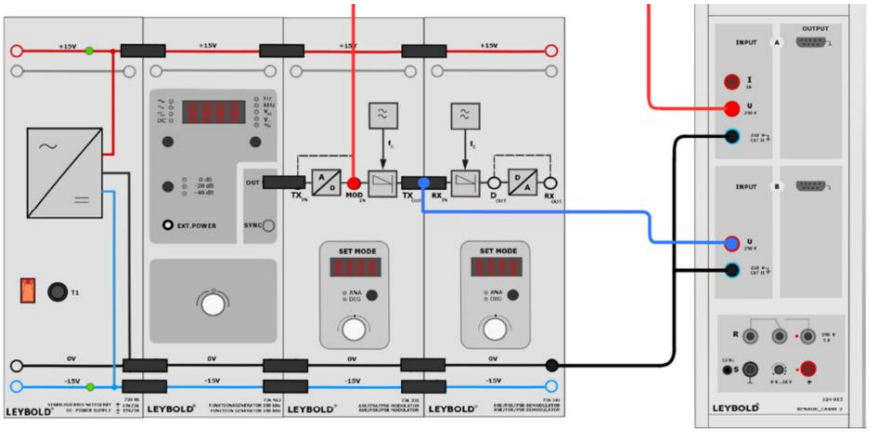


Figure 5: ASK Synchronous Demodulation.[5]

Procedure

Part 1, 2: Amplitude shift keying Modulation (Time and frequency domain)

Hard keying:



The components are assembled as shown in the figure. The function generator is set to produce a square wave with a frequency of 1000 Hz, a peak-to-peak voltage of 10 V, and a duty cycle of 50%. The modulator is configured in digital (DIG) mode, and amplitude shift keying (ASK) with hard keying is selected, using a modulation index of 100% (A_{11}). The Cassy sensor UA1 is connected to the modulating signal, while the Cassy sensor UB1 is connected to the unipolar square-wave signal at the MODIN socket.

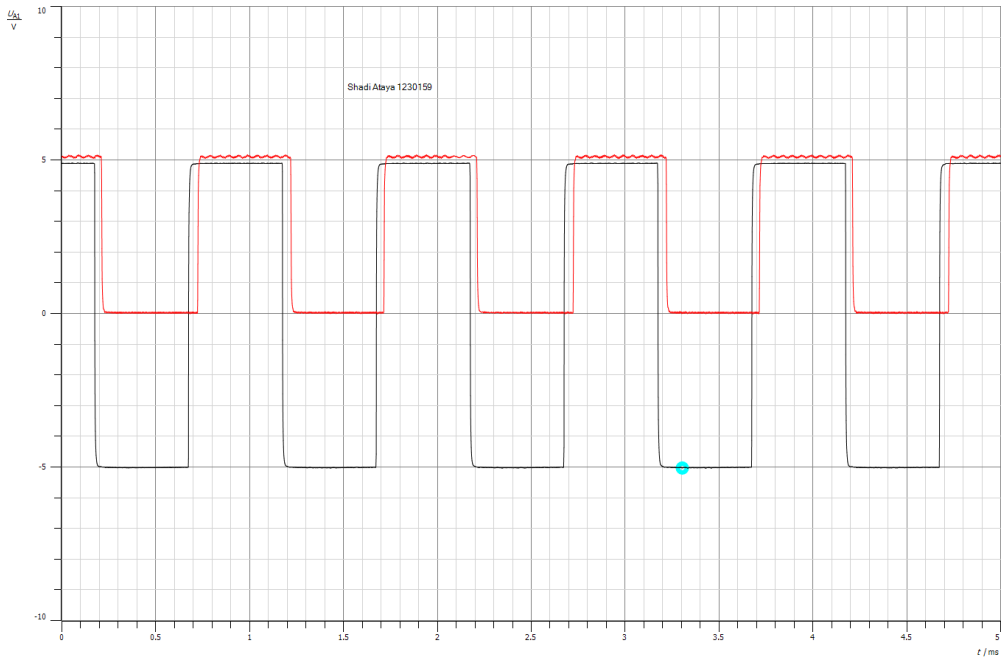


Figure 6: Message signal in time domain.

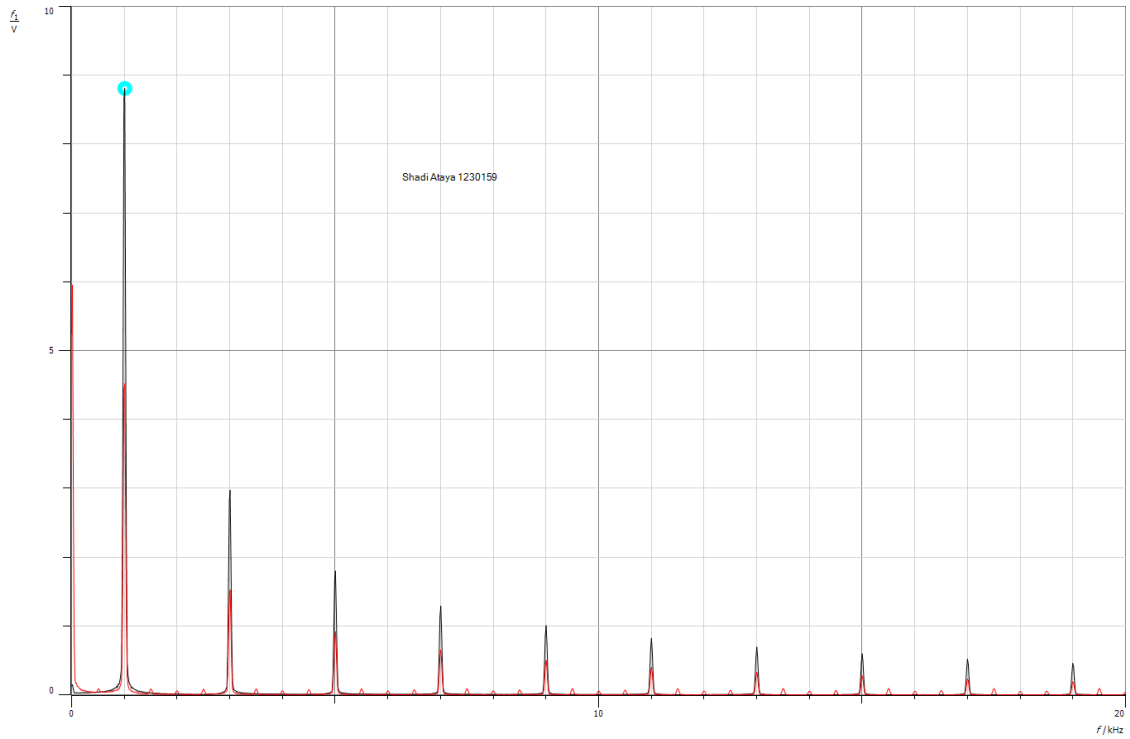


Figure 7: Message signal in frequency domain.

Now, the Cassy sensor UA1 is connected to the unipolar square-wave signal at the MODIN socket, while the Cassy sensor UB1 is connected to the ASK signal at the modulator output TXOUT.

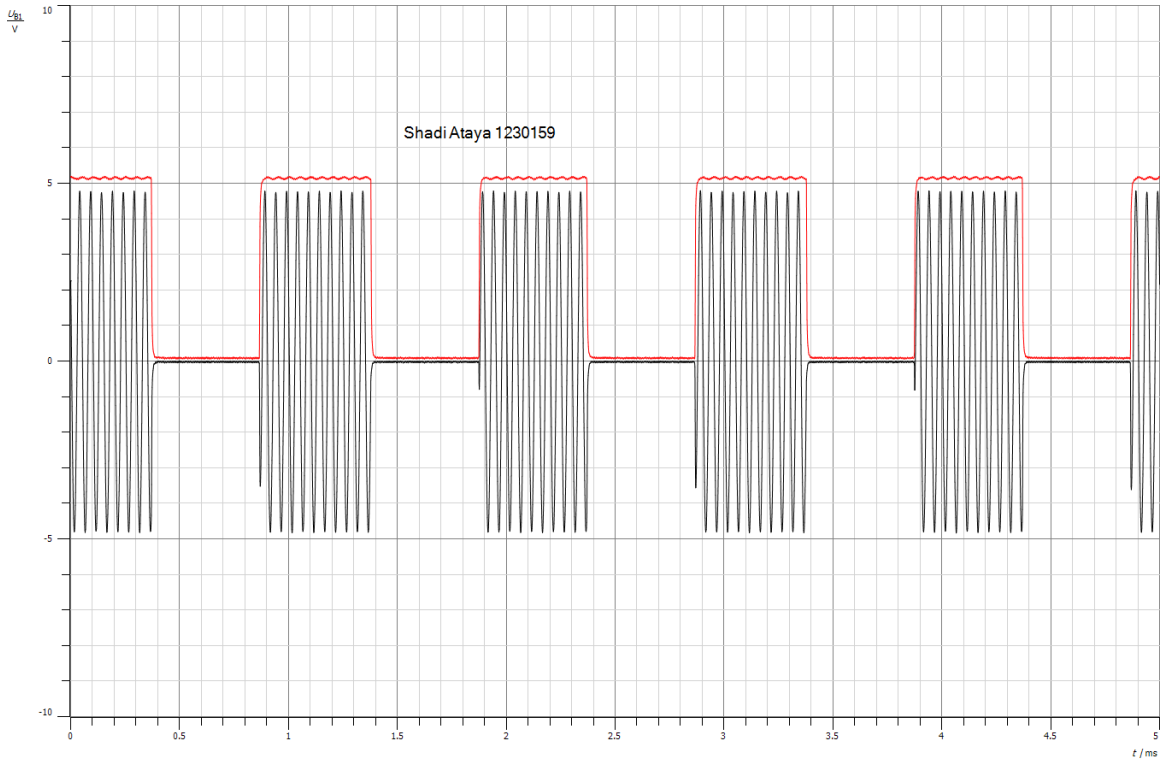


Figure 8: ASK modulation Hard keying with m = 100% (A_11) in time domain

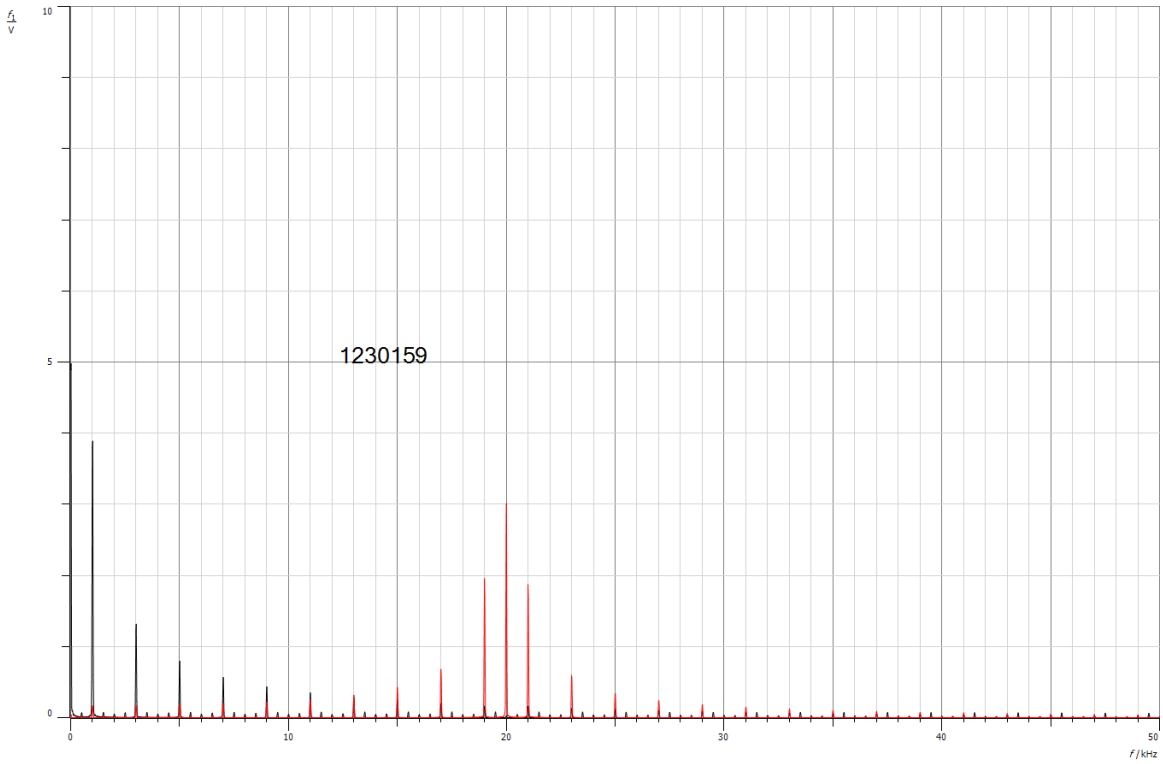


Figure 9: ASK modulation Hard keying with m = 100% (A_11) in frequency domain

- In hard keying ASK modulation (A11) with a 100% modulation index, the time-domain signal exhibits abrupt amplitude transitions, where the carrier is fully present for a logic "1" and completely suppressed for a logic "0", resulting in no distinction between zero-amplitude intervals and the absence of the message signal; this confirms hard keying behaviour and leads to faster switching compared to lower modulation indices since only two amplitude levels are used. In the frequency domain, the spectrum consists of discrete impulses at the carrier frequency and its harmonics, with numerous low-power components caused by the sharp transitions in the time domain. Based on these results, it can be concluded that the message signal is unipolar NRZ, as the carrier is directly switched on and off by the message levels.

The previous part was repeated, but with selecting the ASK - Hard keying with m= 50% (A_21)

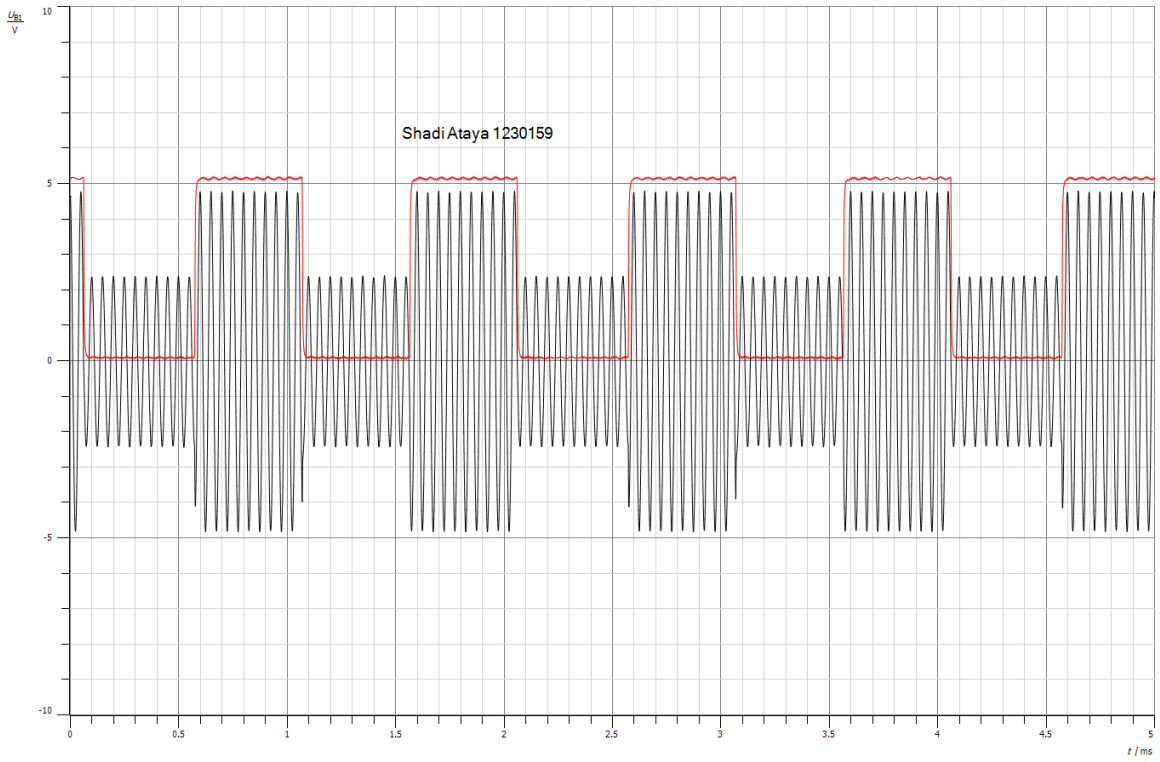


Figure 10: ASK modulation Hard keying with m = 50% (A_21) in time domain

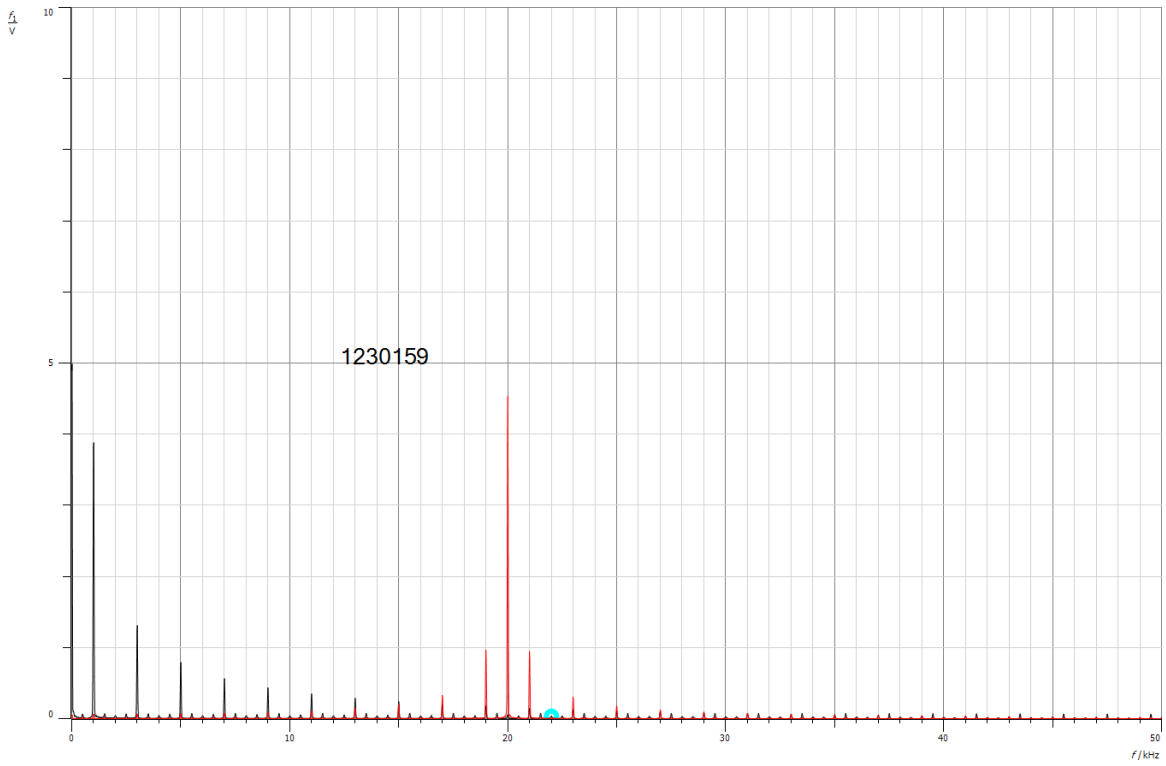


Figure 11: ASK modulation Hard keying with m = 50% (A_21) in frequency domain

- In hard keying ASK modulation (A21) with a 50% modulation index, the time-domain signal still exhibits abrupt amplitude transitions characteristic of hard keying; however, unlike the 100% case, the carrier is not completely suppressed during the low level of the message signal, resulting in two distinct nonzero amplitude levels. This means that when the message corresponds to logic “0”, a reduced-amplitude carrier is still present. In the frequency domain, the spectrum consists of discrete impulses at the carrier frequency and its harmonics, but with fewer visible harmonics and less pronounced low-power components compared to the 100% modulation case. The side impulses are weaker, indicating reduced sideband strength and less overall spectral spreading. This behaviour reflects the direct influence of the modulation index, where a lower value leads to weaker modulation effects and a more concentrated spectral energy distribution around the carrier.

What is the coding method that is used in the experiment system?

Unipolar NRZ

Soft keying:

The connections were kept the same as those used for the hard keying setup. The same procedures were then repeated, but the modulator was configured for amplitude shift keying with soft keying and a modulation index of 100% (A_{12}).

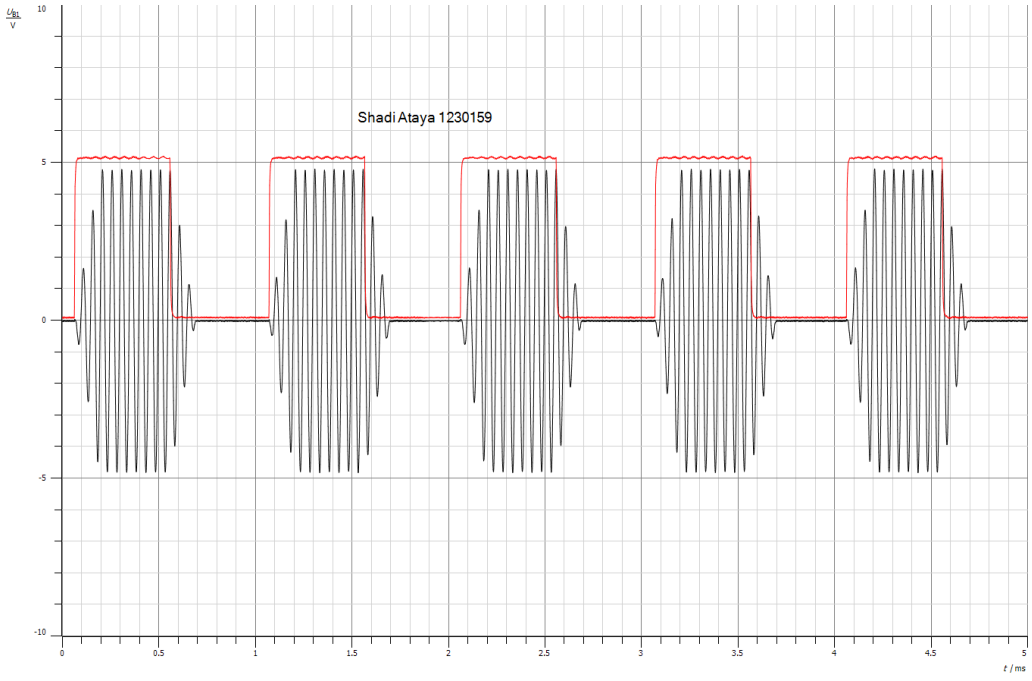


Figure 12: ASK modulation soft keying with m = 100% (A_12) in time domain

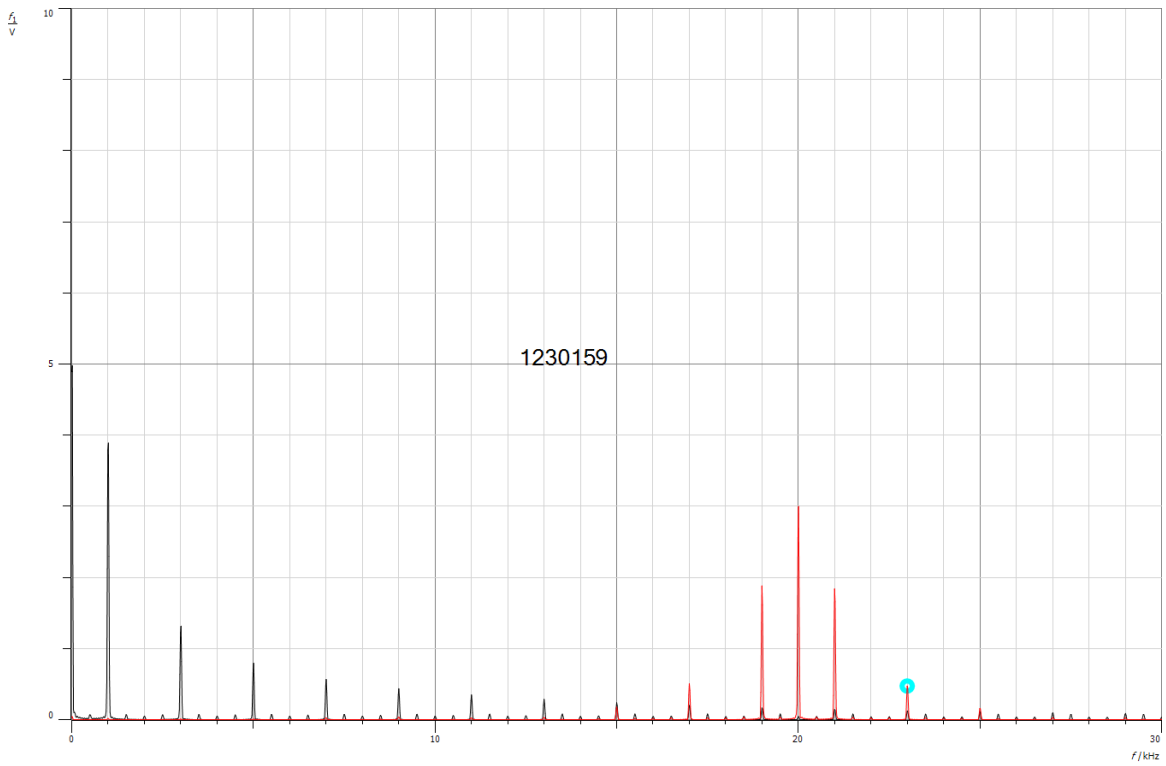


Figure 13: ASK modulation soft keying with m = 100% (A_12) in frequency domain

- With soft keying ASK modulation (A12) at a 100% modulation index, the time-domain waveform exhibits smooth and gradual transitions between amplitude levels rather than the abrupt changes seen in hard keying. This shows that the carrier amplitude varies continuously with the message signal, reducing sudden discontinuities. Consequently, the frequency-domain spectrum contains fewer low-power impulses and less spectral spreading, with most of the signal energy concentrated around the carrier frequency. Compared to hard keying at the same modulation index, soft keying significantly reduces high-frequency components and unwanted harmonics, resulting in a cleaner spectrum and improved bandwidth efficiency.

Finally, the modulator was set to amplitude shift keying with soft keying and a modulation index of 50% (A_{22}), and the same procedure was repeated.

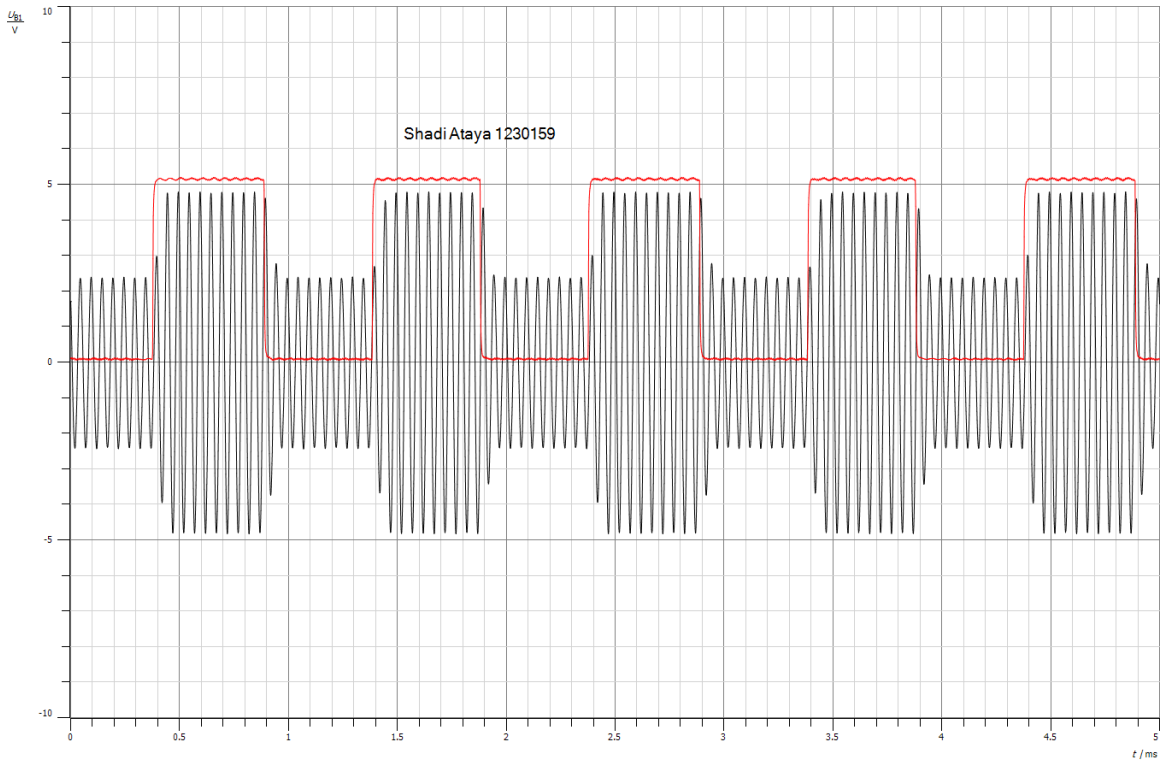


Figure 14: ASK modulation soft keying with m = 50% (A_22) in time domain

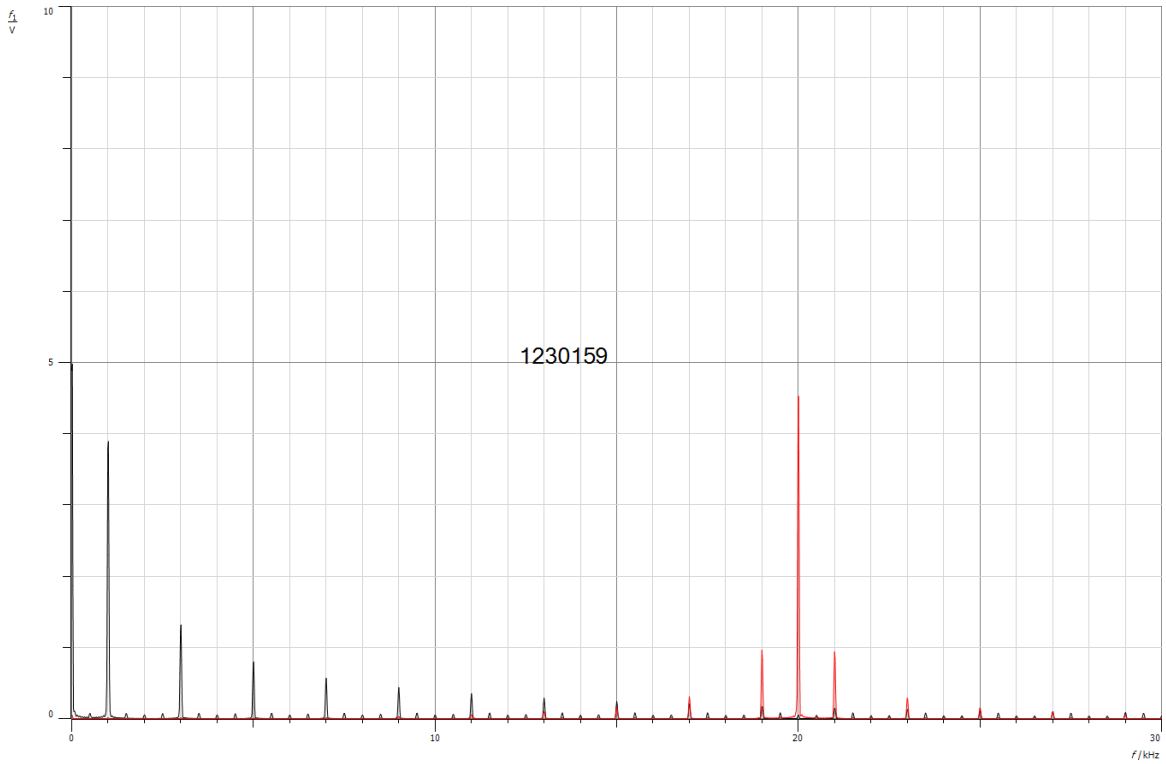


Figure 15: ASK modulation soft keying with m = 50% (A_22) in frequency domain

- With soft keying ASK modulation (A22) at a 50% modulation index, the time-domain signal exhibits smooth and gradual amplitude transitions, like soft keying at 100% modulation but with a reduced depth of amplitude variation. The carrier amplitude is not fully suppressed during the low level of the message signal, resulting in two smoothly varying, nonzero amplitude levels. In the frequency domain, the spectrum shows fewer low-power impulses and reduced spectral spreading compared to hard keying cases, with sidebands that are weaker due to the lower modulation index. Compared to soft keying at 100% modulation, the spectral components are less pronounced, reflecting the direct influence of the reduced modulation depth while maintaining the bandwidth efficiency advantages of soft keying.

Are there any similarities between the amplitude modulation and the amplitude shift keying?

As observed from the previous results, there are clear similarities between amplitude modulation (AM) and amplitude shift keying (ASK). In both techniques, information is conveyed by varying the amplitude of a sinusoidal carrier signal while the carrier frequency remains constant. Both AM and ASK generate sidebands around the carrier frequency in the frequency domain, and the modulation index controls the depth of amplitude variation, directly affecting sideband strength and bandwidth. In the time domain, both signals exhibit an envelope that represents the modulating signal, meaning that ASK can be viewed as an amplitude-modulated carrier driven by a digital waveform.

What do you expect the differences between the hard and soft keying techniques can be?

Feature	Hard	Soft
Bandwidth	**wider spectrum** due to sharp amplitude transitions, which introduce additional high-frequency components.	**narrower spectrum** because smooth transitions reduce high-frequency components.

Detecting the comparator threshold:

In this part, the two thresholds (upper and lower) that switch the ASK signal from the high amplitude to the low amplitude were determined. The same connections used in the previous parts were kept unchanged. The function generator was set to a DC signal with a voltage of 0 V, and the modulator was configured for amplitude shift keying using hard keying with a modulation index of 100% (A_{11}). The control voltage was then reduced from 2.5 V until the low-level (zero-volt) output appeared on the ASK signal.

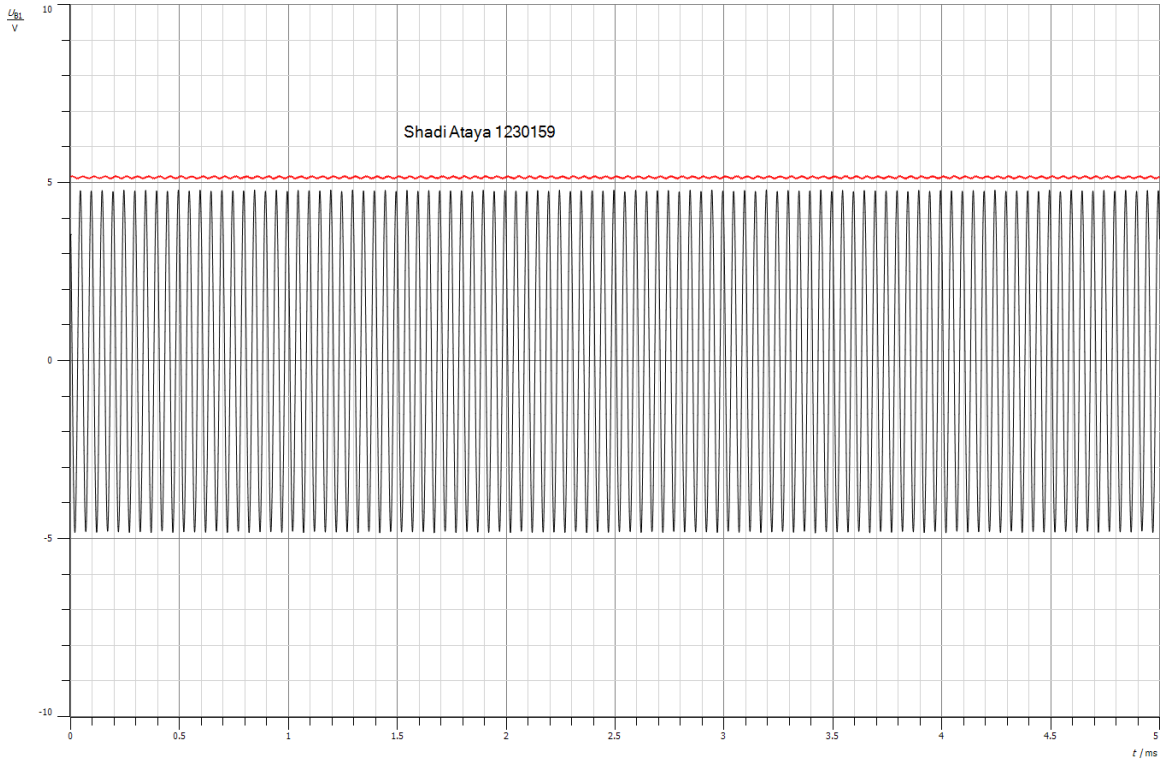


Figure 16: ASK signal at DC upper threshold value in time domain.

DC upper threshold value = 2.4

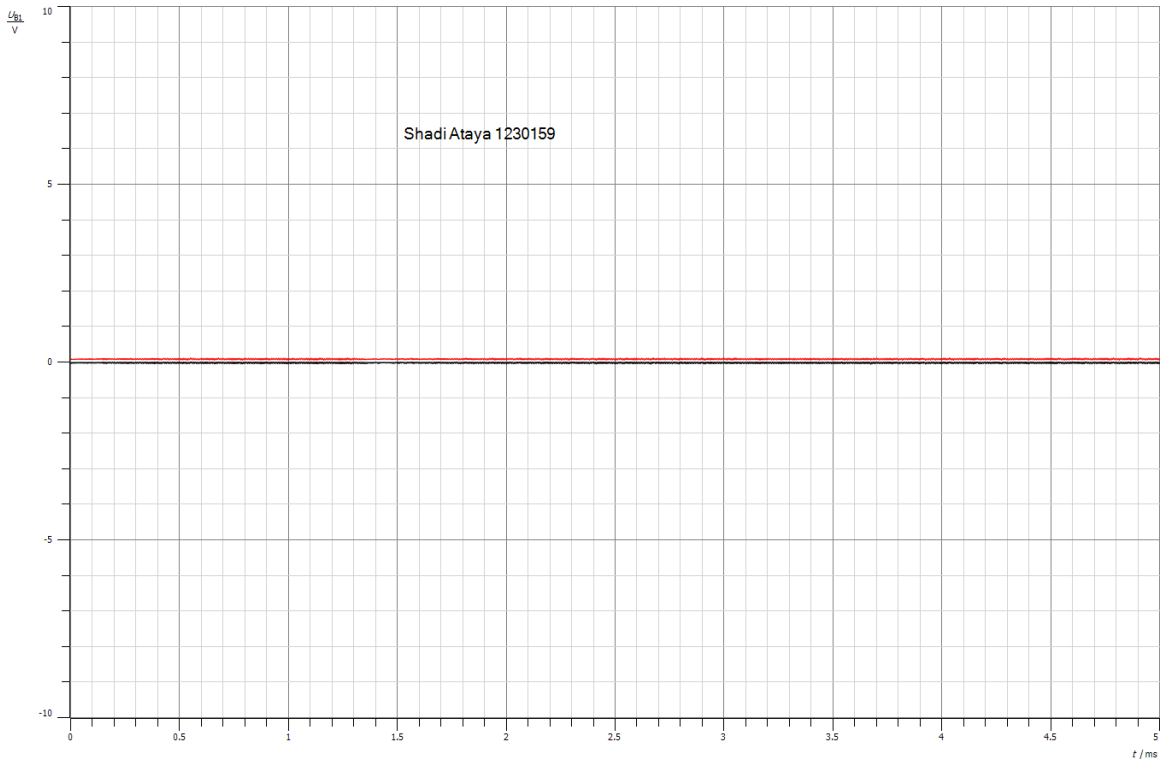


Figure 17: ASK signal at DC lower threshold value in time domain.

DC lower threshold value = 1.8

- As the DC control voltage was gradually increased to determine the upper threshold, the ASK output transitioned from the low-amplitude state to the high-amplitude state as expected. Conversely, when the DC voltage was reduced to identify the lower threshold and reached 0 V, the ASK signal was completely suppressed and disappeared. This observation indicates that below the lower threshold the modulator turns the carrier fully off, causing the ASK signal to cease switching between high and low amplitude levels.

Studying the effect of changing the message signal amplitude (Time and frequency domains):

The same connections as in the previous parts were kept unchanged, and the message signal amplitude was reduced to 1.5 V, corresponding to a peak-to-peak voltage of 3 V.

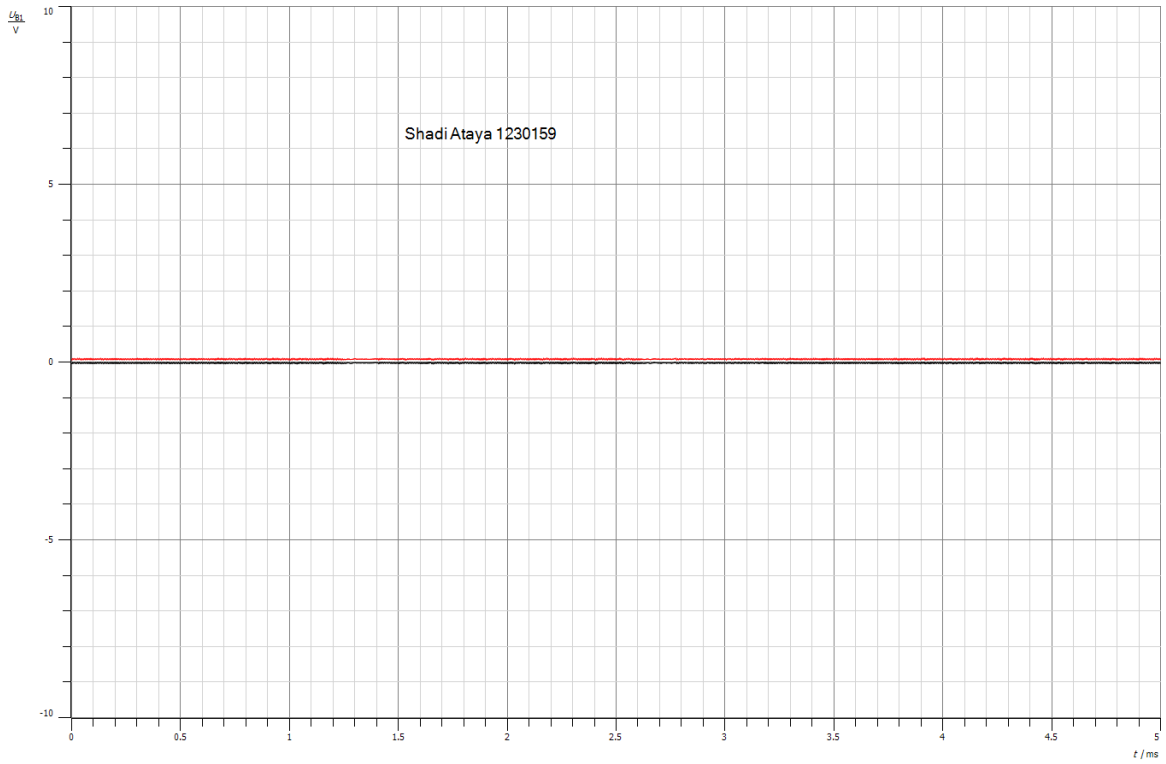


Figure 18: ASK signal with amplitude = 1.5 volts in time domain

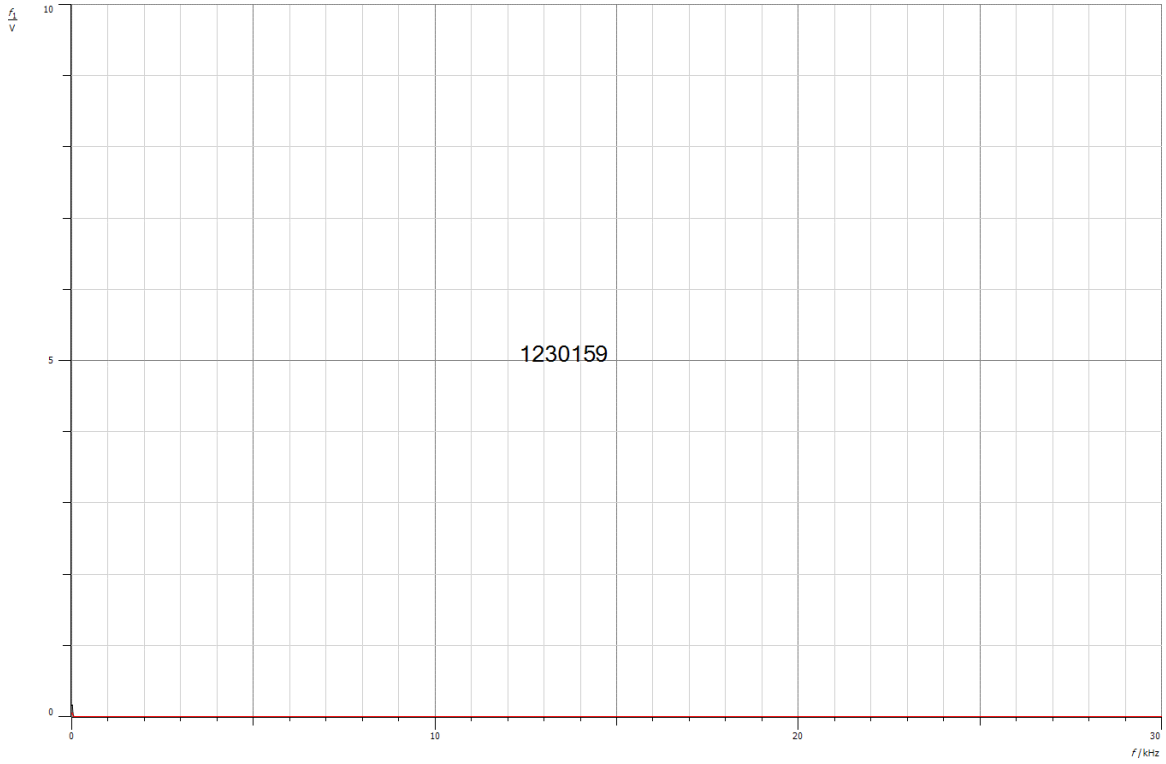


Figure 19: ASK signal with amplitude = 1.5 volts in frequency domain

- The message signal amplitude directly influences the ASK spectrum. When the message amplitude is high, the carrier undergoes larger amplitude variations, producing stronger and more pronounced sidebands and resulting in a broader spectrum. As the message amplitude is reduced, the amplitude variation of the carrier decreases, causing the sidebands to weaken and gradually disappear, which leads to a narrower spectrum. In this case, when the message amplitude was set to 1.5 V, which is below the lower threshold value of 1.8 V, the carrier was completely suppressed, causing the ASK signal and its sidebands to disappear entirely. This observation confirms that the message amplitude controls the modulation depth in ASK and directly affects sideband strength, bandwidth usage, and the overall spectral characteristics of the signal.

Studying the effect of changing the message signal frequency on (Time and frequency domains):

The same connections as in the previous parts were kept unchanged. The function generator was set to generate a square wave with a frequency of 500 Hz, a peak-to-peak voltage of 10 V, and a duty cycle of 50%. The Cassy sensor UA1 was connected to the unipolar square-wave signal at the MODIN socket, and the Cassy sensor UB1 was connected to the ASK signal at the TXOUT output of the modulator.

- **(A_ 11)**

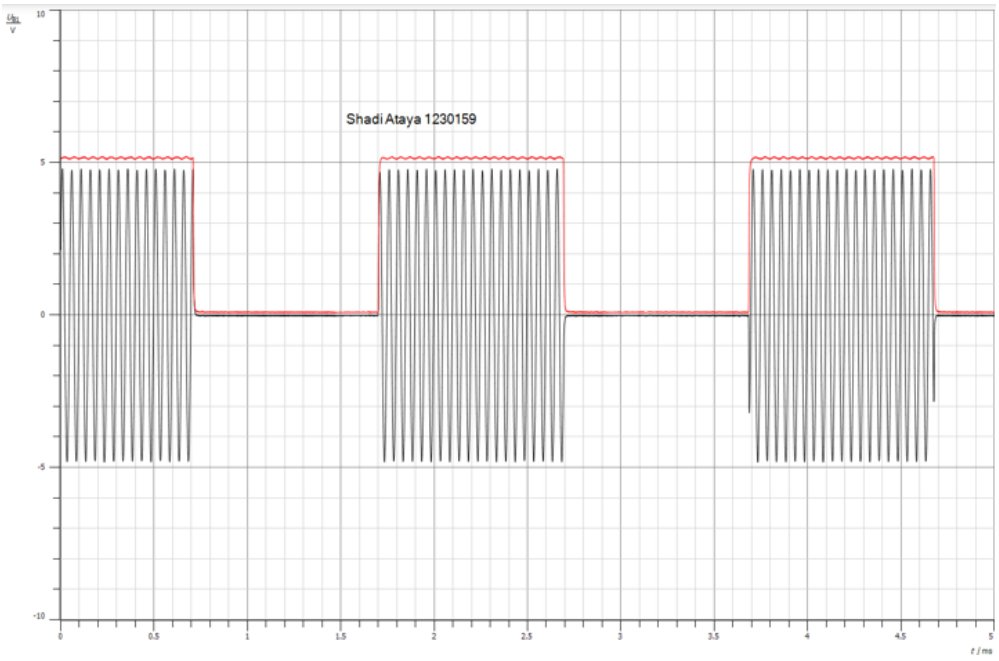


Figure 20: ASK modulated signal with frequency = 500 Hz (A_11) in time domain

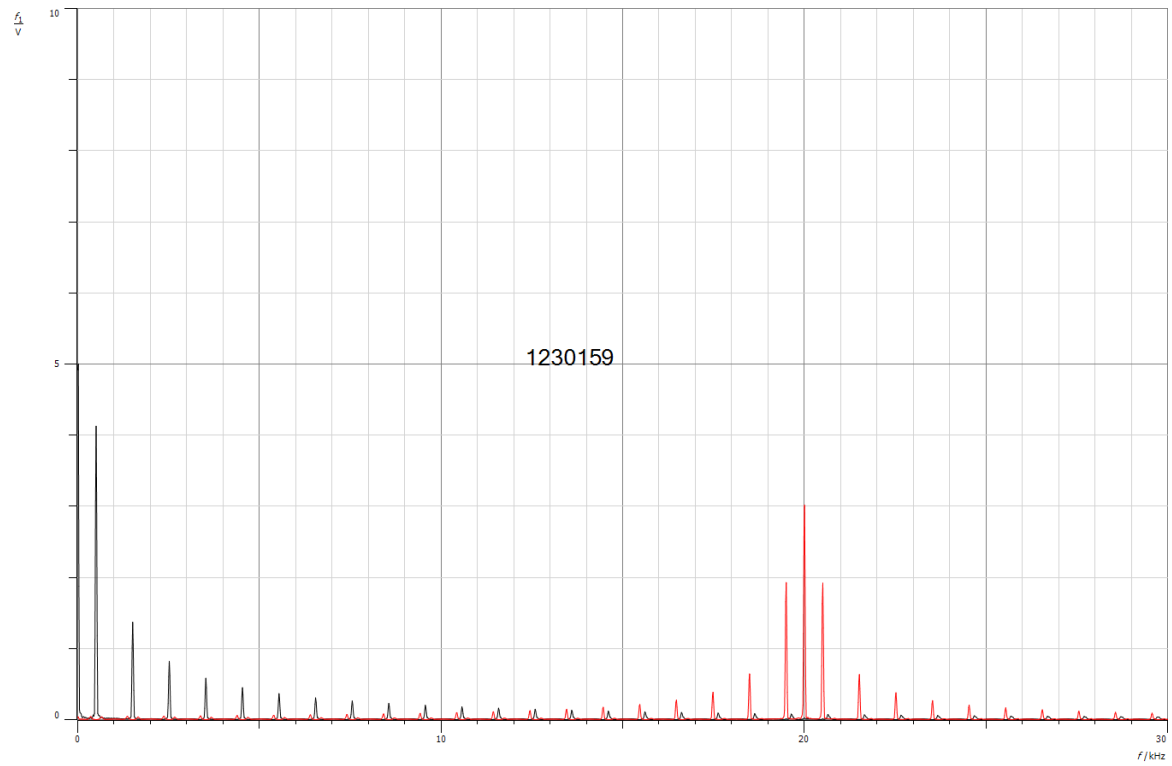


Figure 21: ASK modulated signal with frequency = 500 Hz (A_11) in frequency domain

- **(A_21)**

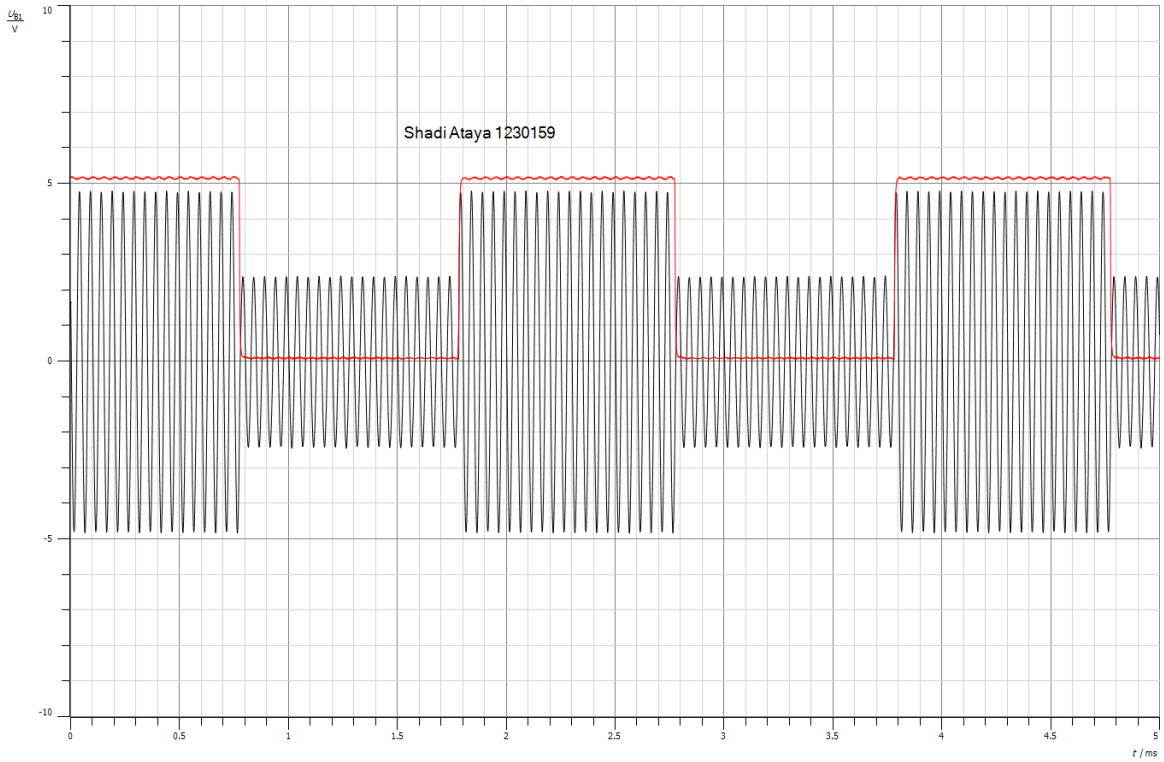


Figure 22: ASK modulated signal with frequency = 500 Hz (A_21) in time domain

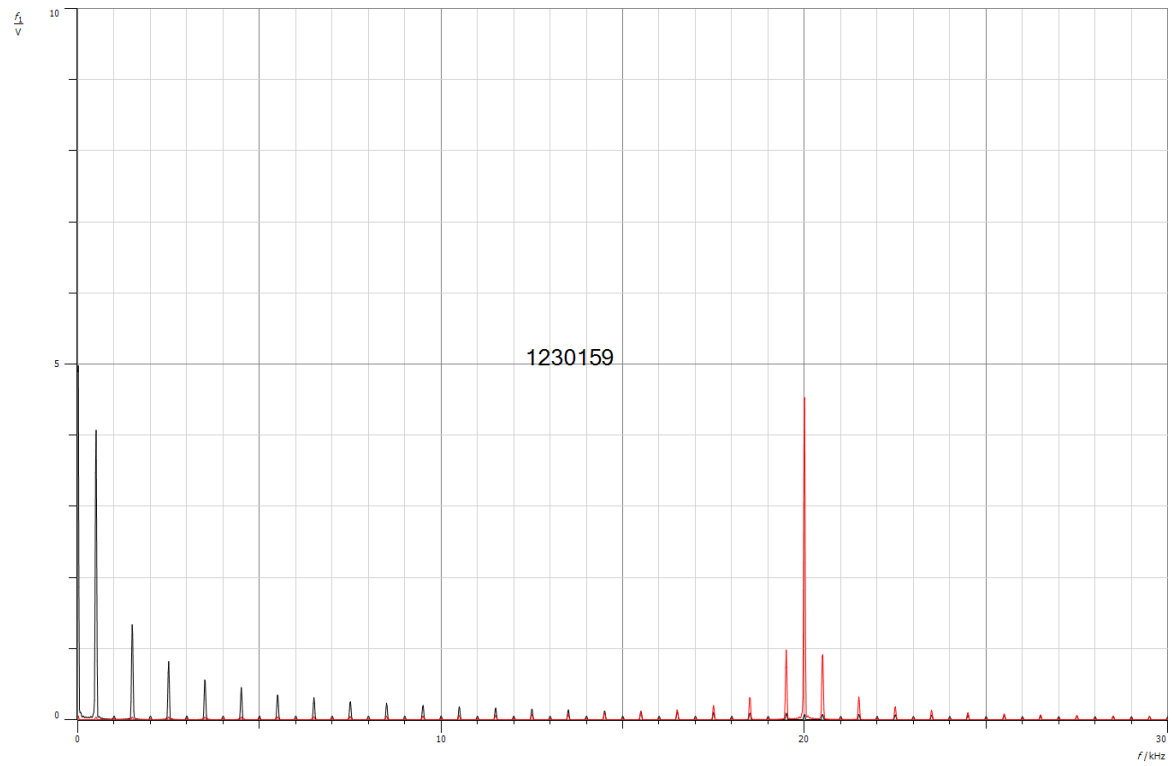


Figure 23: ASK modulated signal with frequency = 500 Hz (A_21) in frequency domain

- **(A_12)**

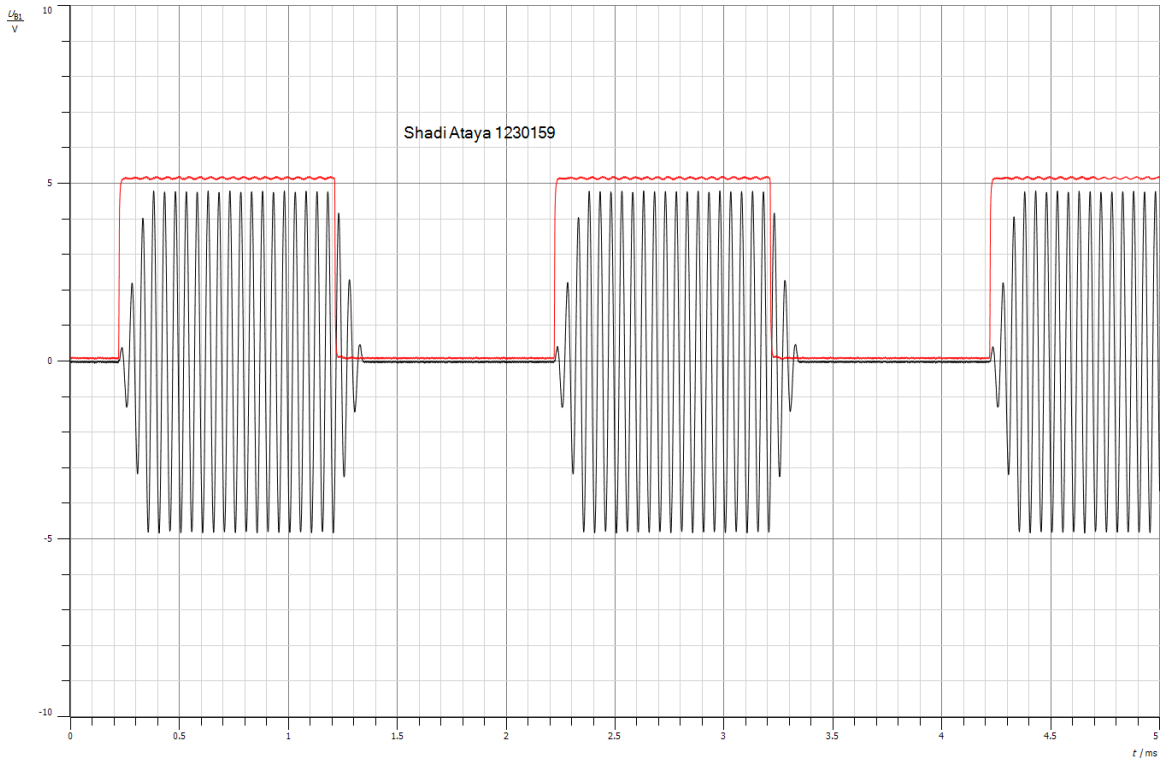


Figure 24: ASK modulated signal with frequency = 500 Hz (A_12) in time domain

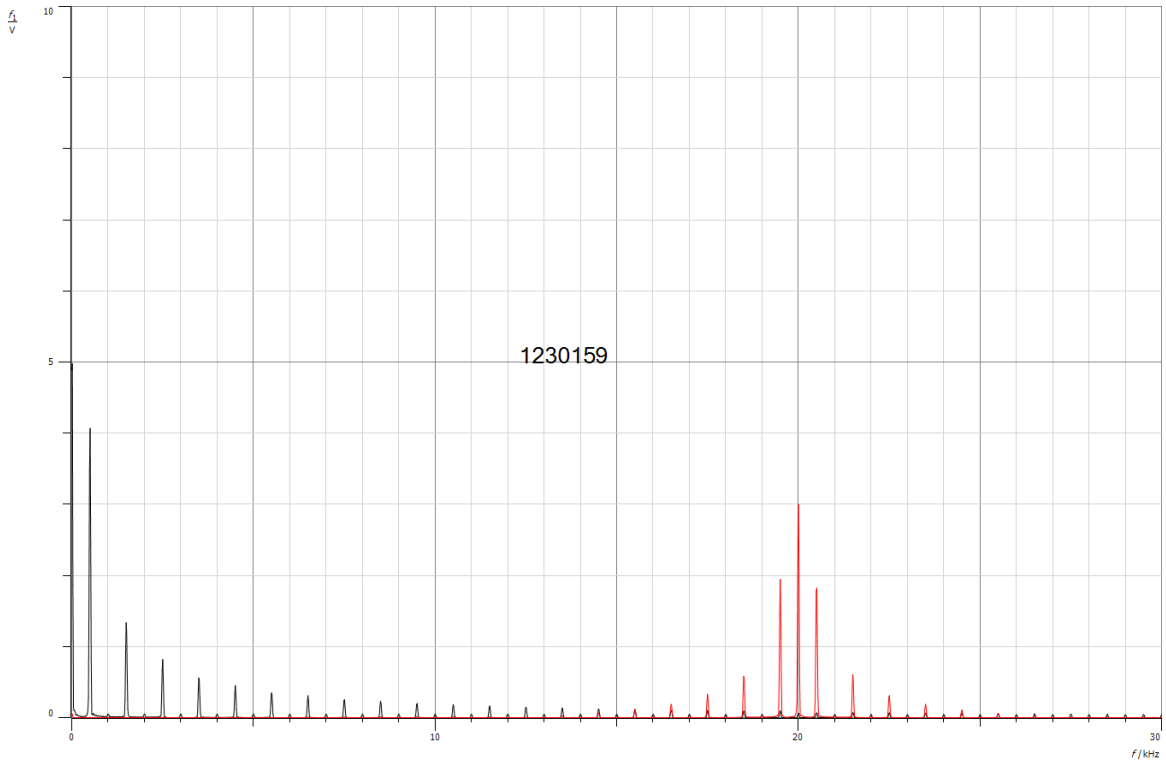


Figure 25: ASK modulated signal with frequency = 500 Hz (A_12) in frequency domain

- **(A_22)**

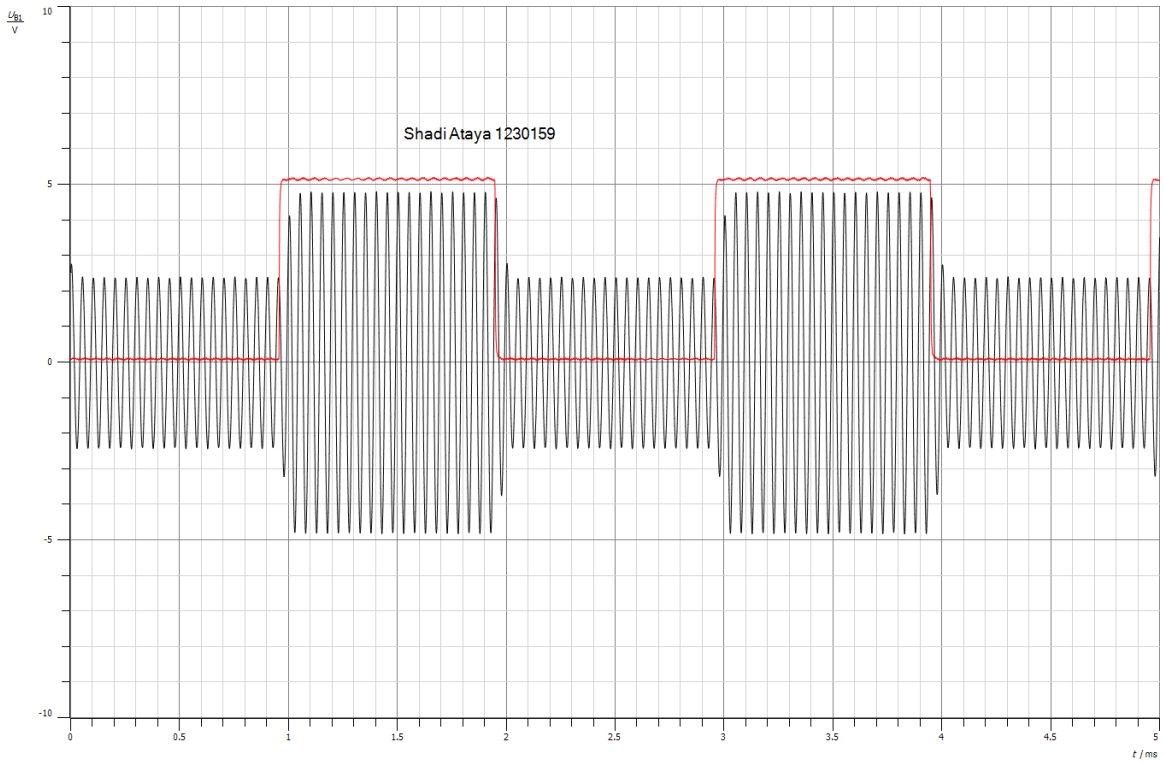


Figure 26: ASK modulated signal with frequency = 500 Hz (A_22) in time domain

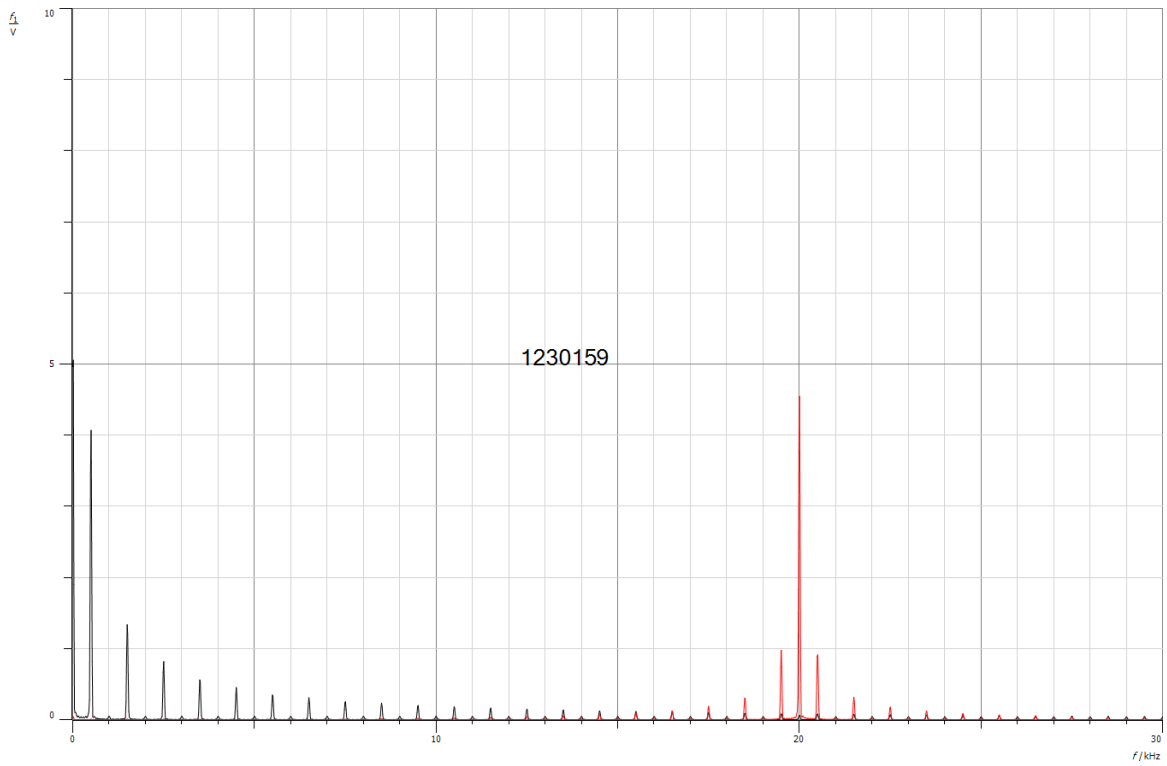


Figure 27: ASK modulated signal with frequency = 500 Hz (A_22) in frequency domain

- The spectrum of an ASK signal is directly influenced by the message frequency. Low message frequencies produce sidebands that are close to the carrier frequency, resulting in a narrower spectrum and lower bandwidth. As the message frequency increases, the sidebands move farther from the carrier, broadening the spectrum and increasing bandwidth. Faster transitions in the message signal also introduce higher-frequency components, further widening the spectrum. Therefore, the ASK signal's bandwidth is closely tied to the frequency of the message signal: higher message frequencies require a wider spectral range to represent the signal accurately.

Studying the effect of changing the message signal duty cycle (Time and frequency domains):

The same connections as in the previous part were kept unchanged. The function generator was set to produce a square wave with a frequency of 1000 Hz, a peak-to-peak voltage of 10 V, and a duty cycle of 10%. The duty cycle was then gradually increased, and its effect on the ASK spectrum was observed.

(A_11)

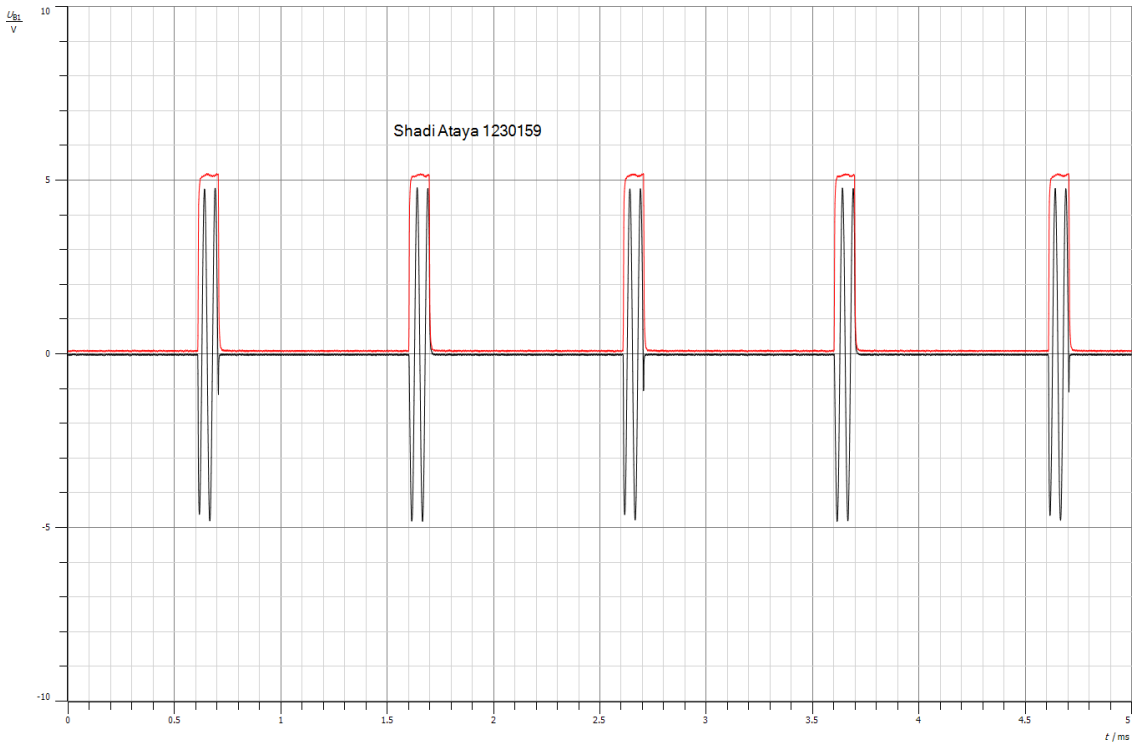


Figure 28: ASK modulated signal with D = 10% (A_11) in time domain

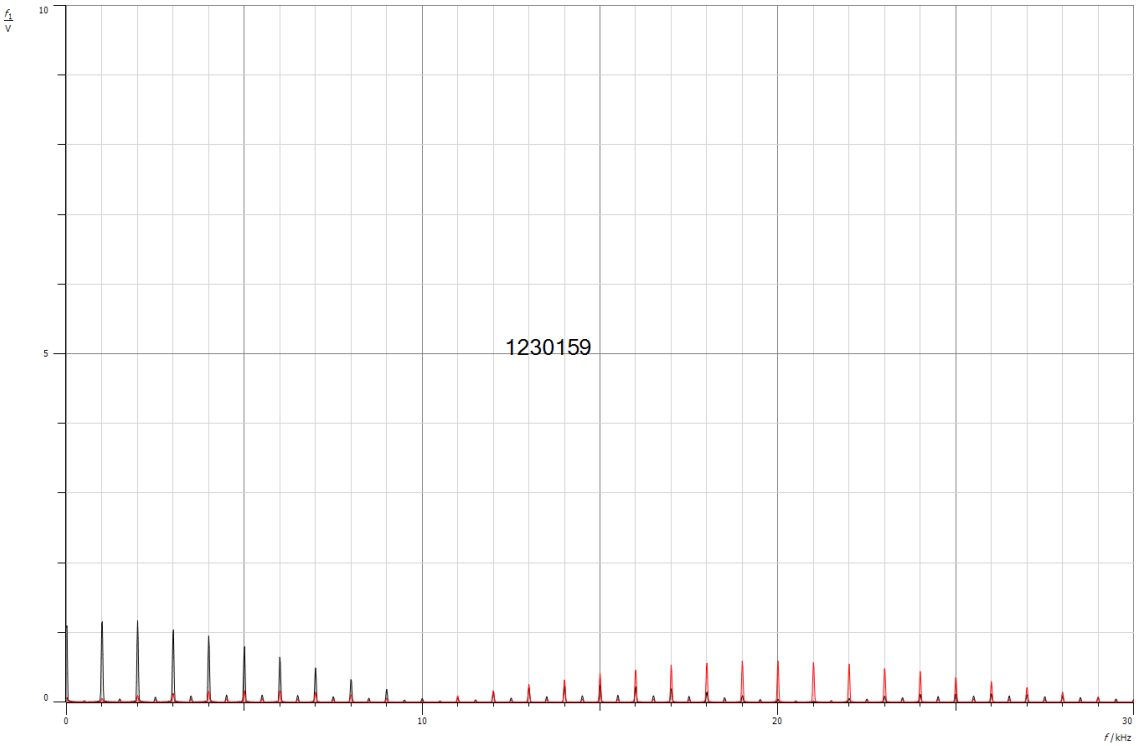


Figure 29: ASK modulated signal with D = 10% (A_11) in frequency domain

(A_21)

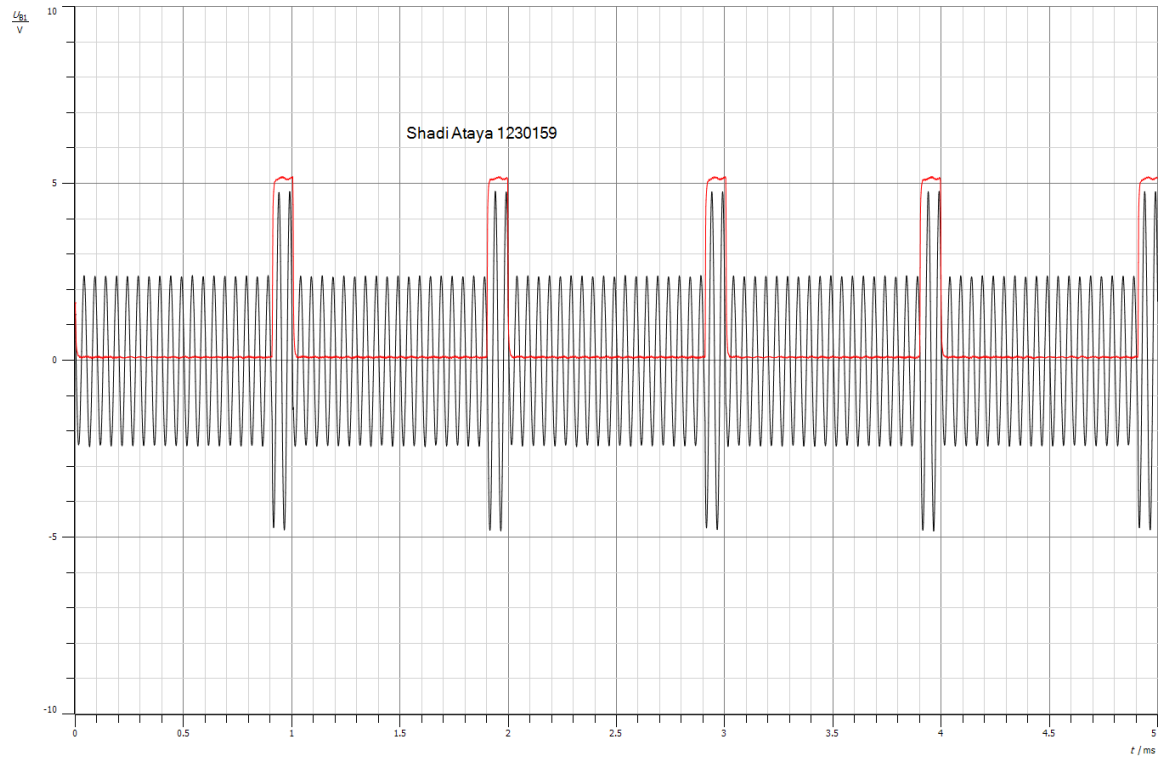


Figure 30: ASK modulated signal with D = 10% (A_21) in time domain

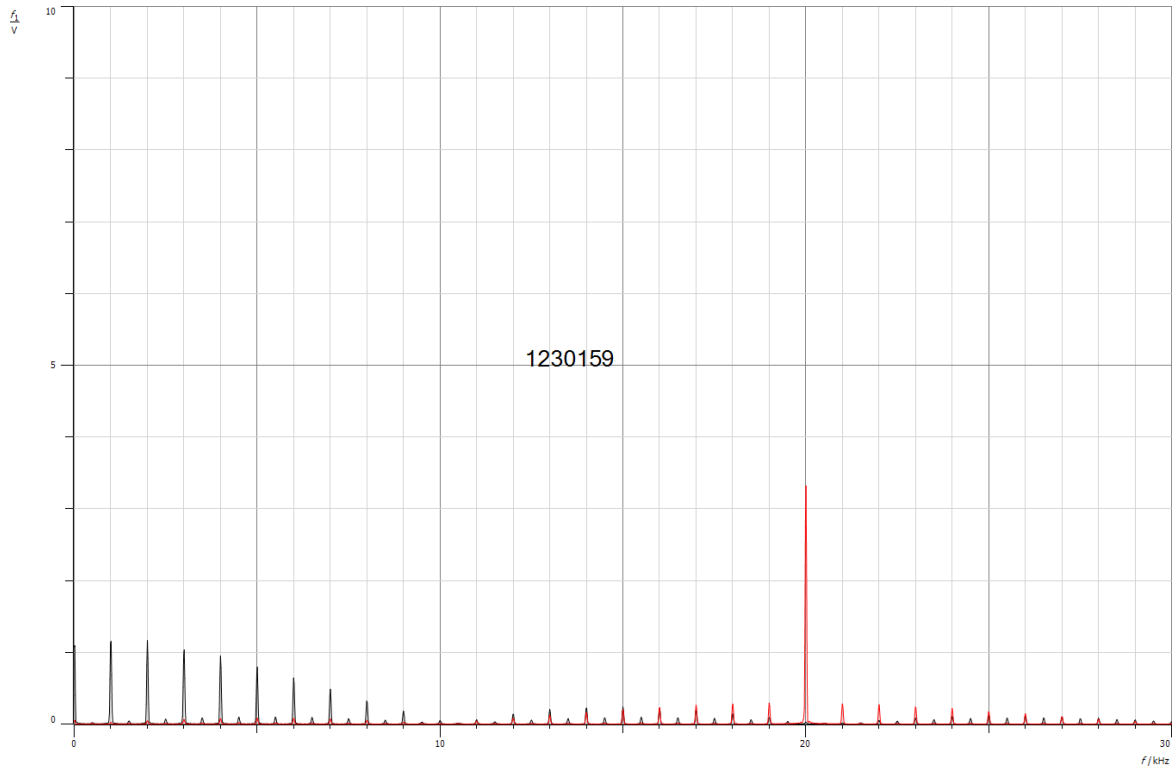


Figure 31: ASK modulated signal with D = 10% (A_21) in frequency domain

(A_12)

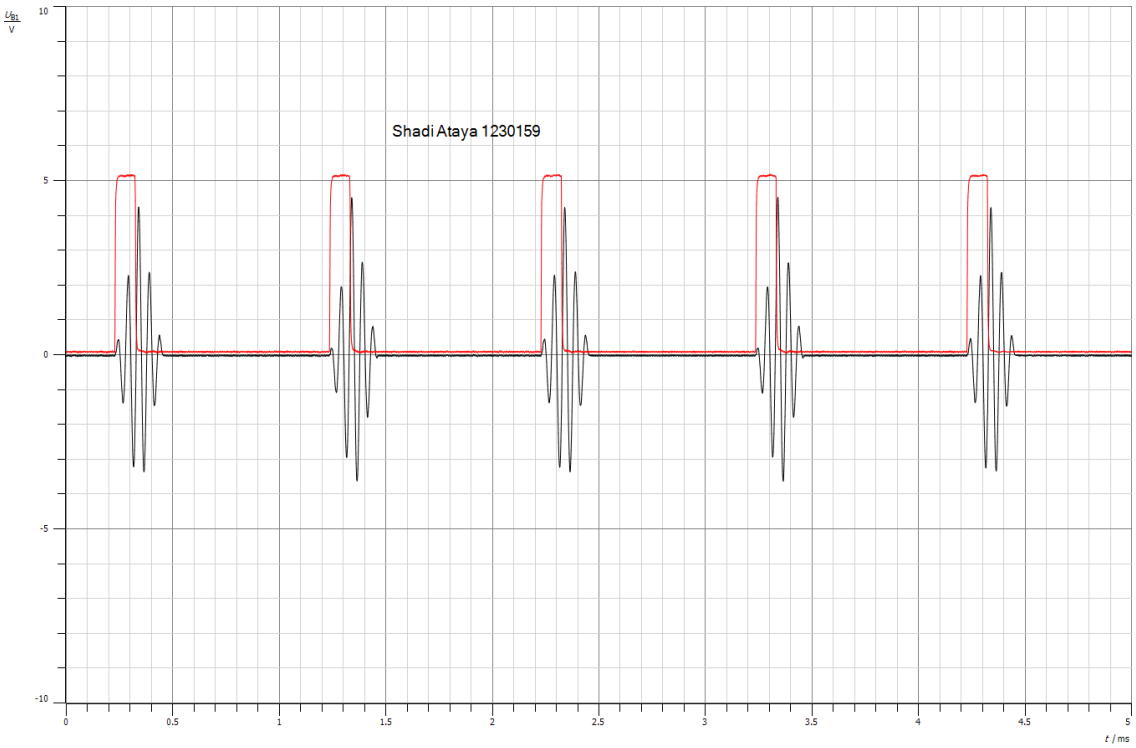


Figure 32: ASK modulated signal with D = 10% (A_12) in time domain

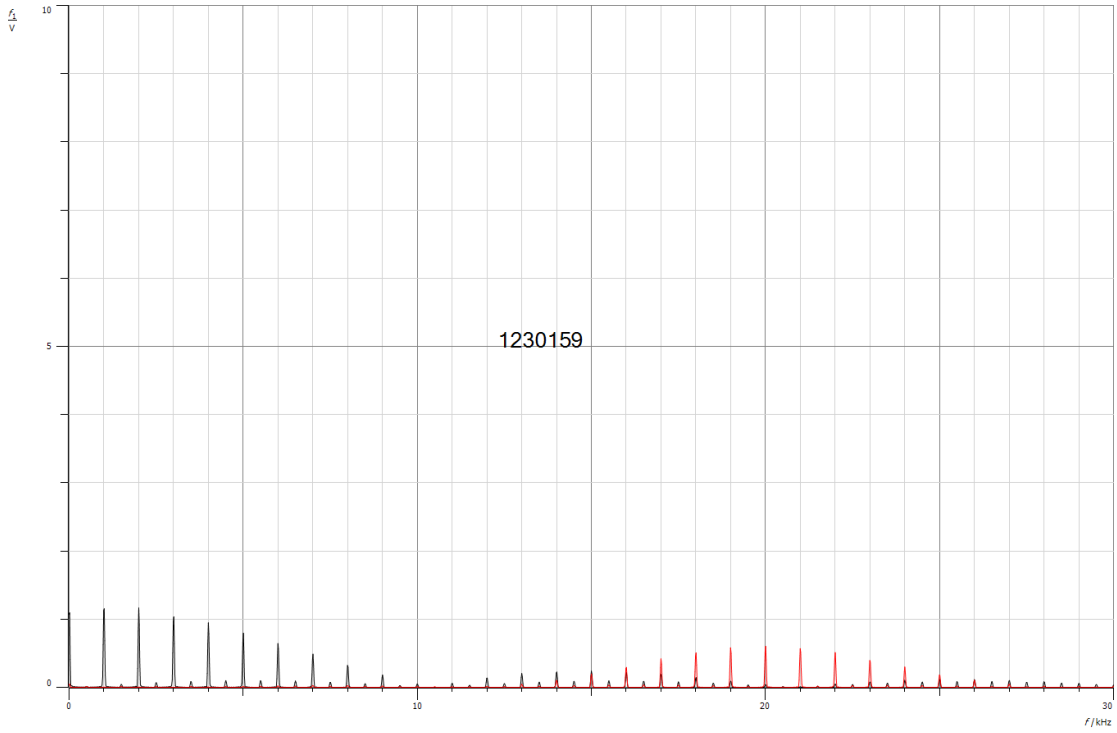


Figure 33: ASK modulated signal with D = 10% (A_12) in frequency domain

(A_22)

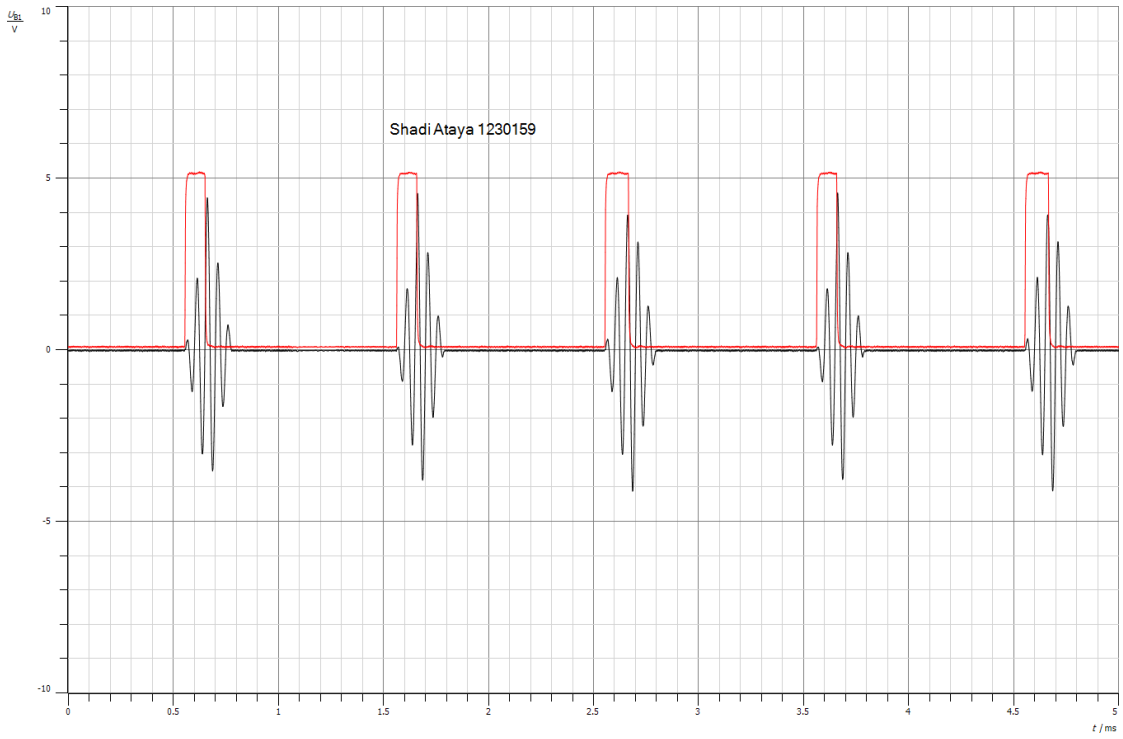


Figure 34: ASK modulated signal with D = 10% (A_22) in time domain

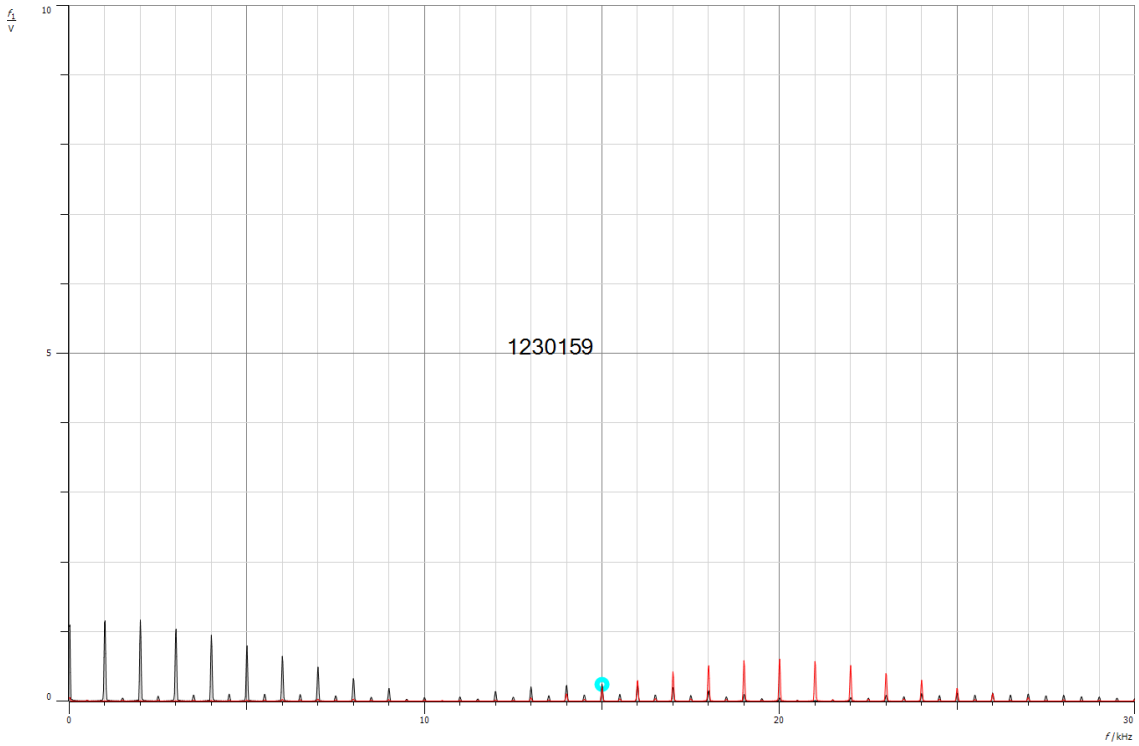
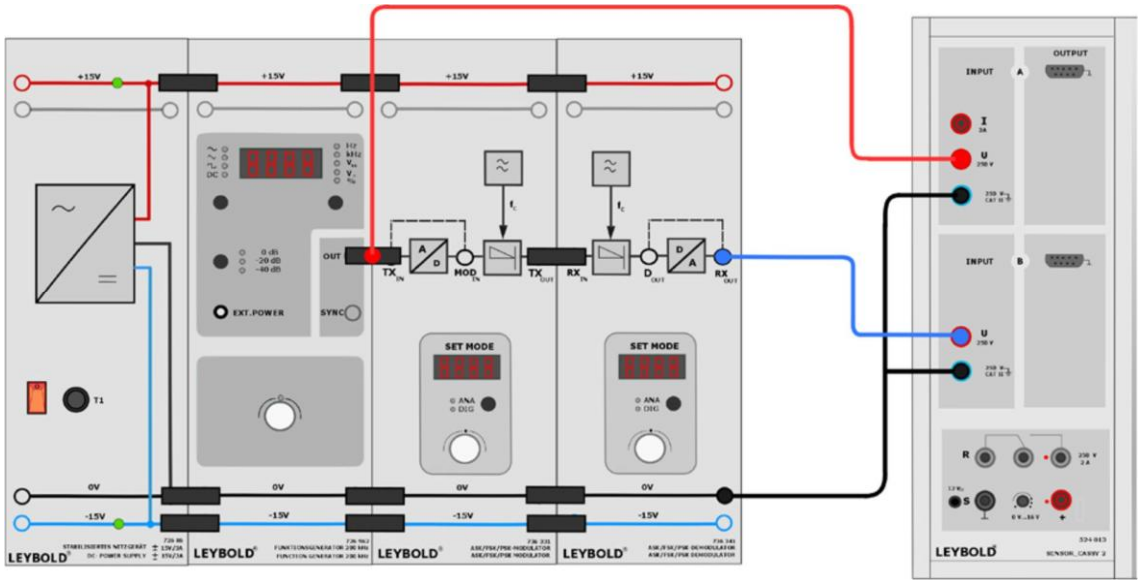


Figure 35: ASK modulated signal with D = 10% (A_22) in frequency domain

- The message duty cycle significantly impacts the ASK spectrum. Increasing it broadens the spectrum with stronger sidebands, while decreasing it narrows the spectrum with weaker sidebands. This aligns with modulation principles, where duty cycle variations affect modulation depth and spectral characteristics. Adjusting the duty cycle allows for precise control over bandwidth usage and signal quality in ASK modulation.

Part 3: Amplitude shift keying Demodulation (Time and frequency domains)



The components were assembled as shown in the figure. The function generator was set to produce a square wave with a frequency of 1000 Hz, a peak-to-peak voltage of 10 V, and a duty cycle of 50%. The demodulator was set to digital (DIG) mode, and the demodulation type (A) was selected. The Cassy sensor UA1 was connected to the bipolar message square-wave signal at the MODIN socket, and the Cassy sensor UB1 was connected to the demodulated signal at the RXOUT output.

Hard Keying:

The modulator mode was set to digital (DIG), and amplitude shift keying with hard keying and a modulation index of 100% (A_{11}) was selected then the (A21) was selected.

(A_11)

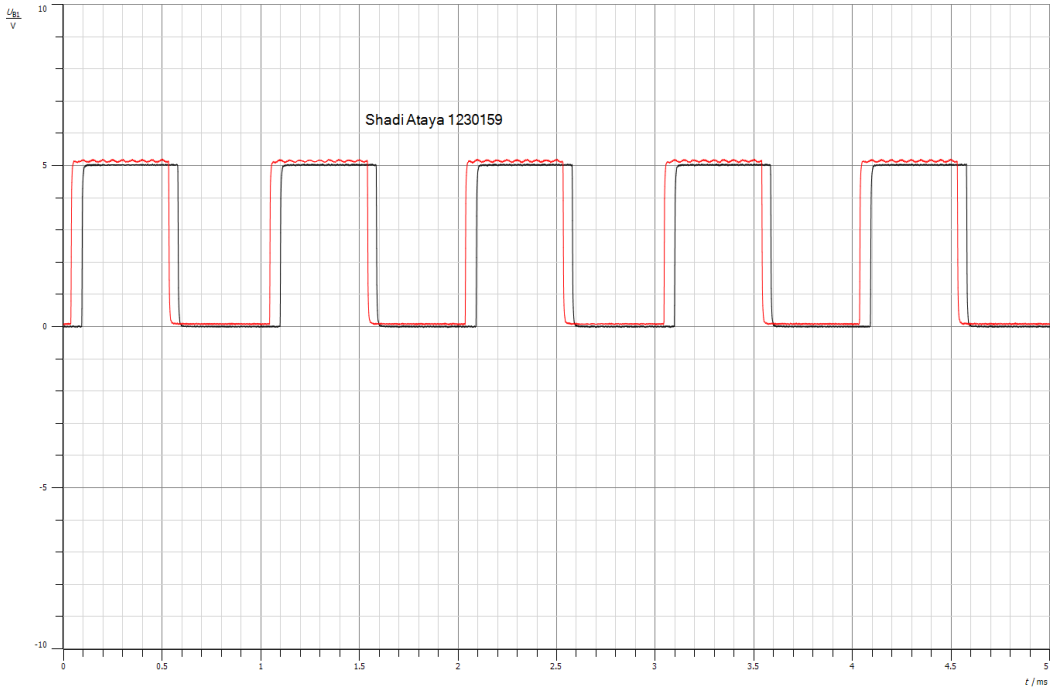


Figure 36: ASK demodulation Hard Keying with m = 100% (A_11) in time domain

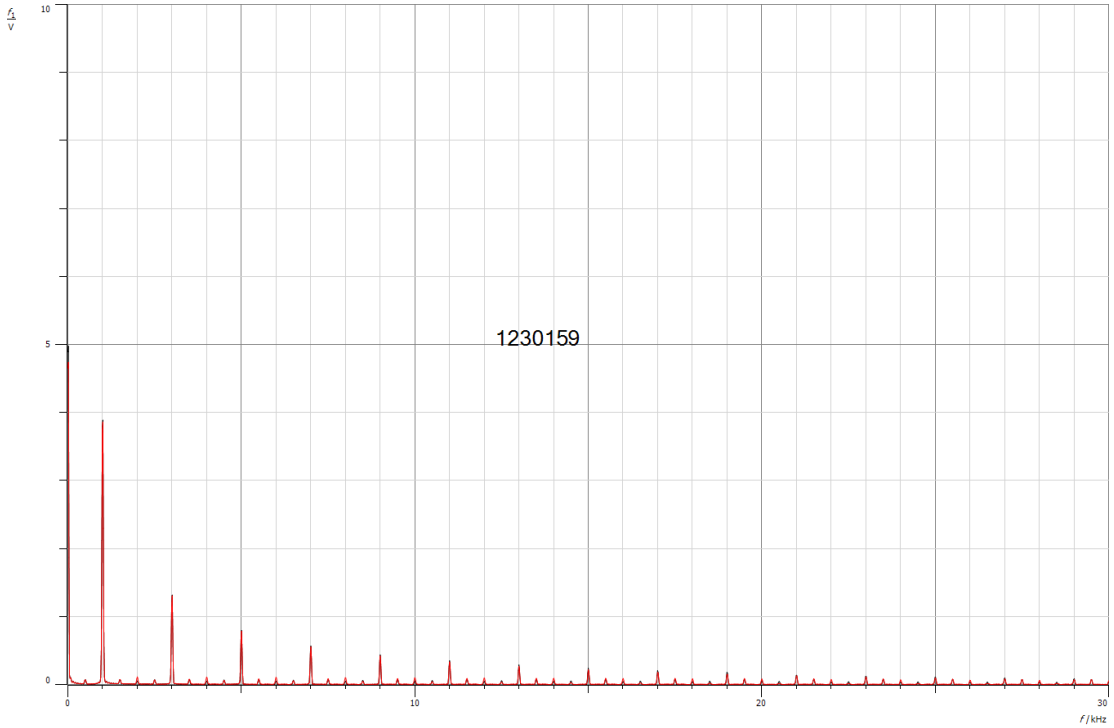


Figure 37: ASK demodulation Hard Keying with m = 100% (A_11) in frequency domain

(A_21)

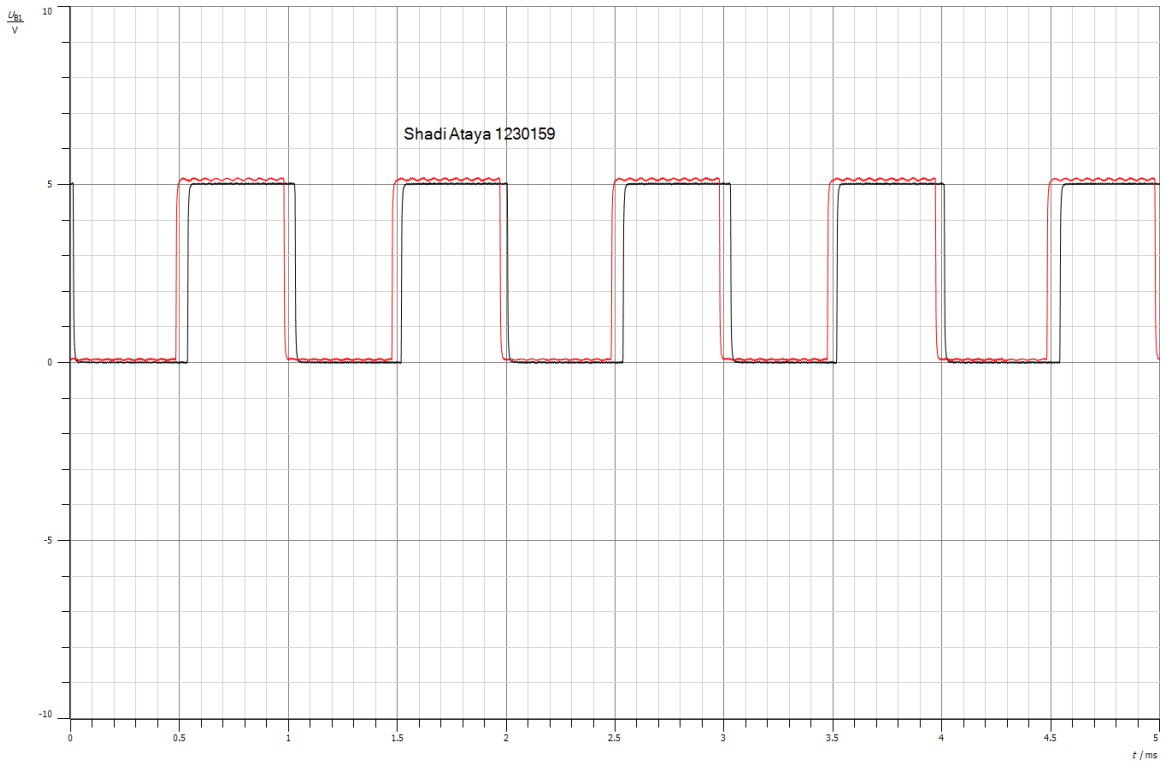


Figure 38: ASK demodulation Hard Keying with m = 50% (A_21) in time domain

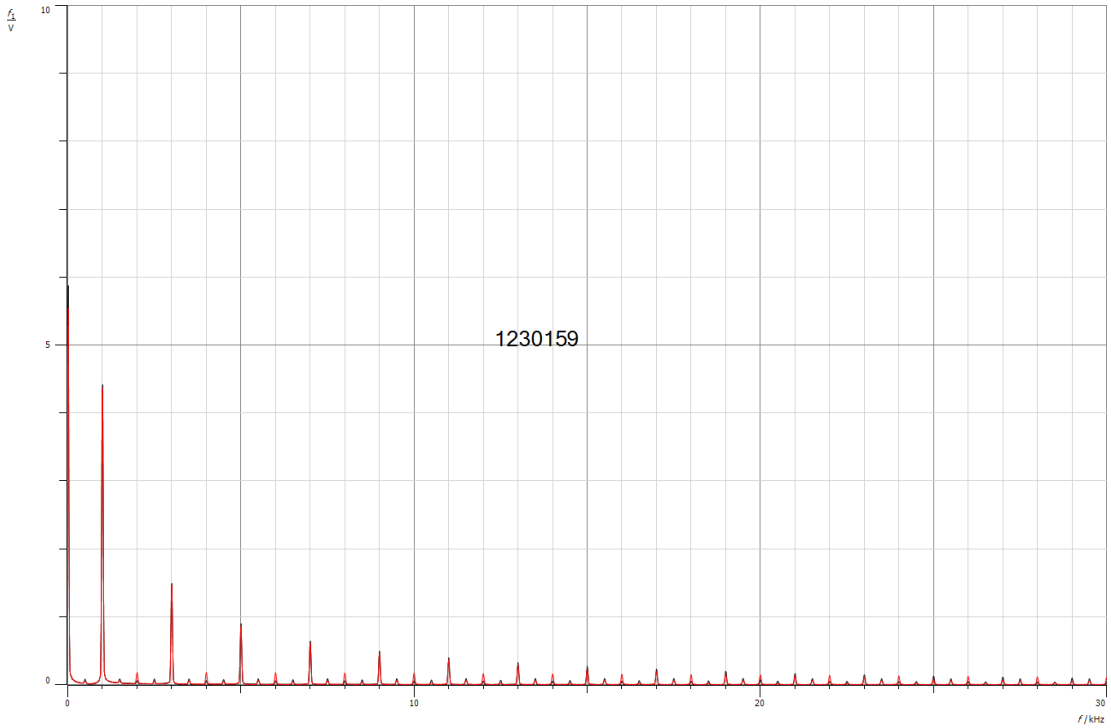


Figure 39: ASK demodulation Hard Keying with m = 50% (A_21) in frequency domain

Does the demodulation of the ASK 100% and 50% hard keying process success?

In ASK, hard keying both 100% and 50% generally allows reliable demodulation because the carrier amplitude changes abruptly, making the presence or absence of the carrier easy to detect. 100% hard keying provides the clearest signal, while 50% hard keying reduces the amplitude swing, slightly increasing sensitivity to noise but still usually demodulates correctly.

Soft Keying:

The connections were kept the same as those used for the hard keying part. The same procedures were repeated, but this time amplitude shift keying with soft keying and a modulation index of 100% (A_{12}) was selected, and then amplitude shift keying with soft keying and a modulation index of 50% (A_{22}) was selected.

(A_12)

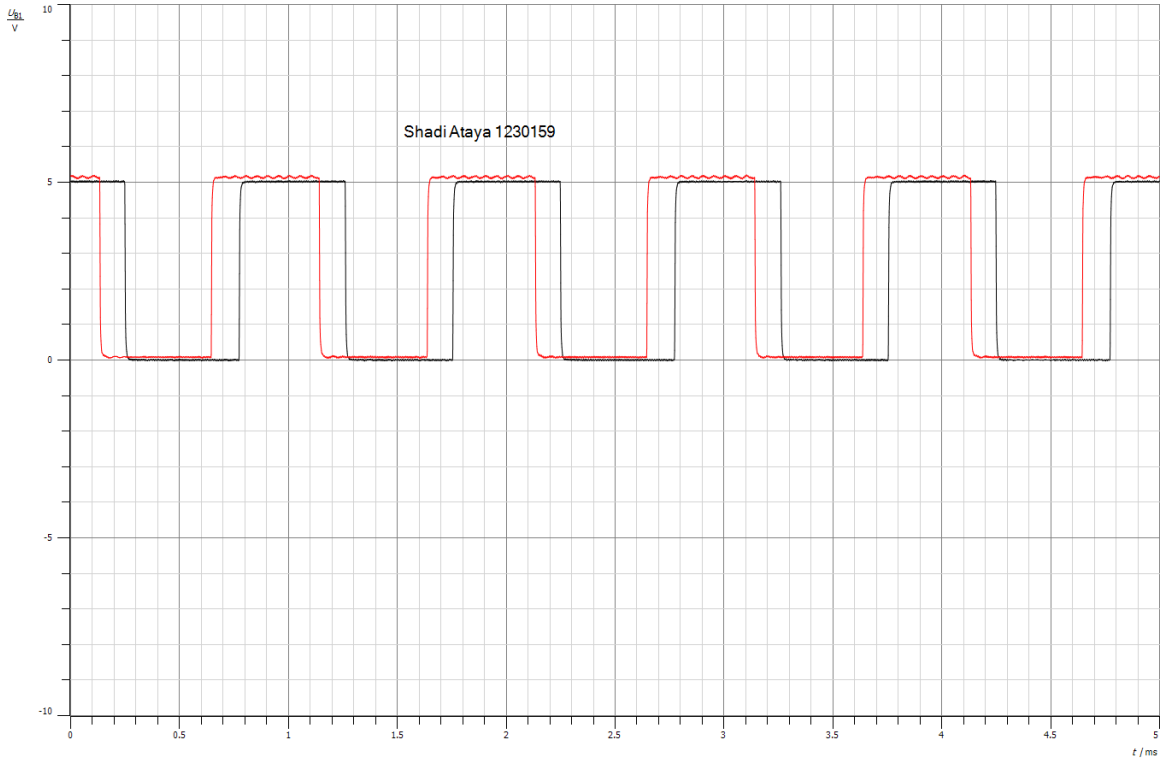


Figure 40: ASK demodulation soft Keying with m = 100% (A_12) in time domain

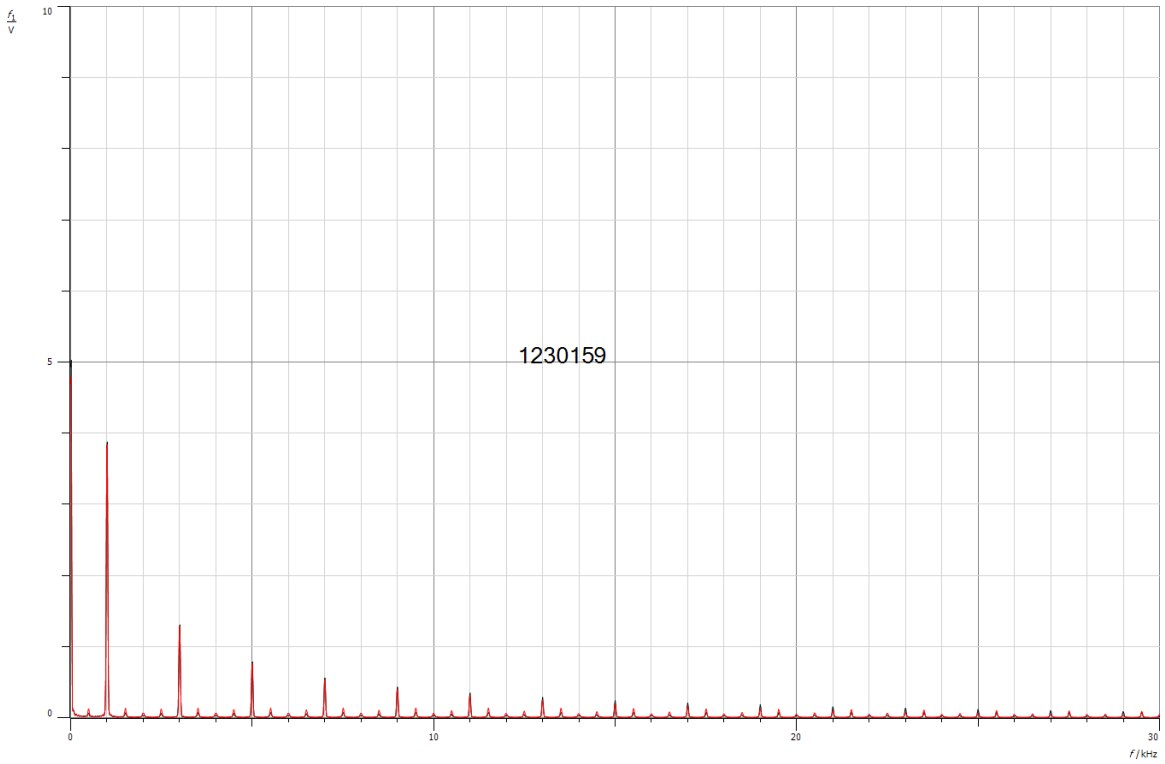


Figure 41: ASK demodulation soft Keying with m = 100% (A_12) in frequency domain

(A_22)

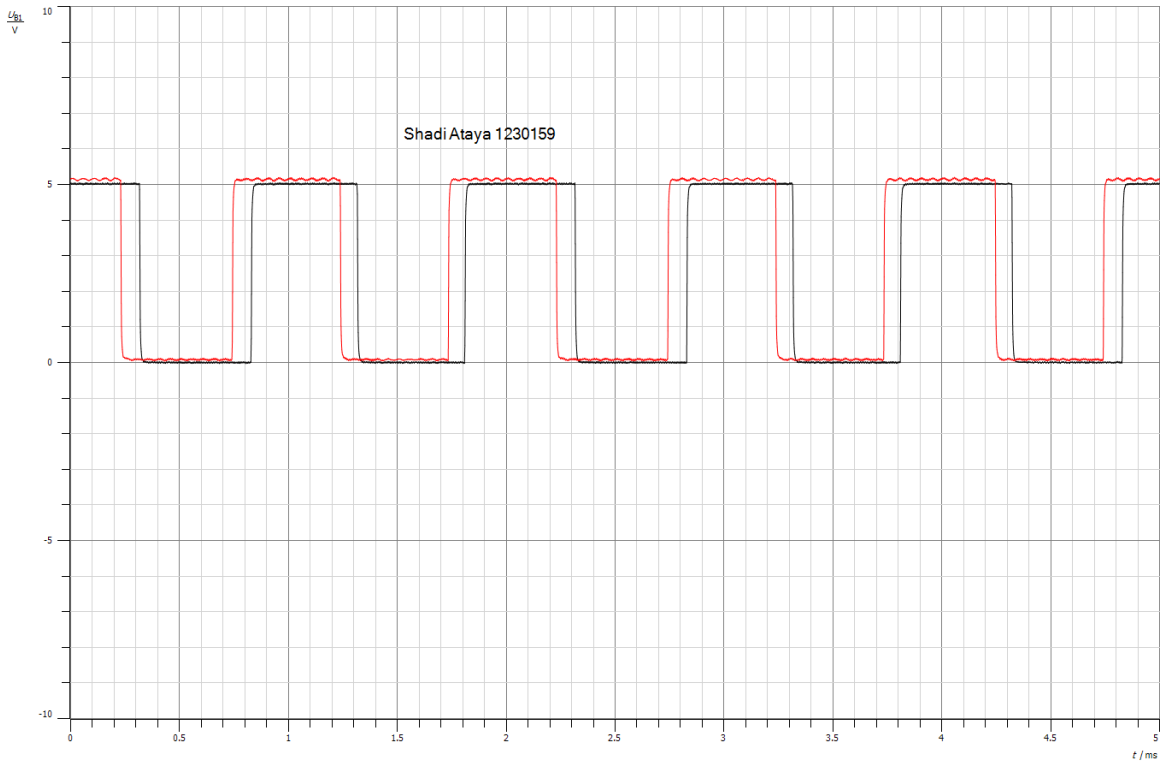


Figure 42: ASK demodulation soft Keying with m = 50% (A_22) in time domain

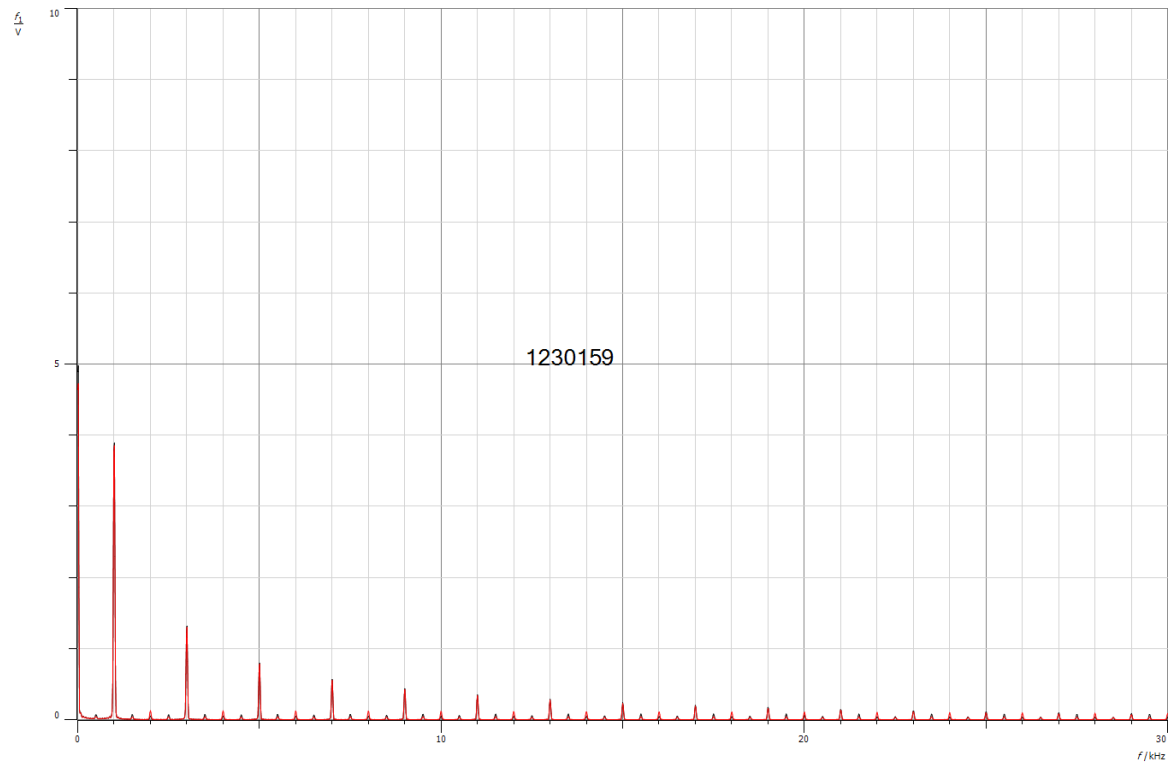


Figure 43: ASK demodulation soft Keying with m = 50% (A_22) in frequency domain

Does the demodulation of the ASK 100% and 50% soft keying process success?

Soft keying gradually changes the carrier amplitude. With 100% soft keying, demodulation usually works, but slower transitions can introduce minor errors. 50% soft keying, which has smaller amplitude swings combined with smooth transitions, makes demodulation less reliable, as noise or channel distortions can more easily affect the recovered signal.

Questions

What is the main difference between the message signal and the demodulated signal?

The message signal is the original baseband signal used to modulate the carrier and is ideally clean and undistorted. The demodulated signal is the recovered version at the receiver. Although it carries the same information, it may have reduced amplitude, rounded edges, or slight distortions compared to the original message.

What causes this difference? And does it consider as a disadvantage?

This difference is caused by the receiver components, mainly the envelope detector and low-pass filter, which introduce delay and smooth the signal, reducing its amplitude. It is generally

considered a disadvantage**,** but a minor one, because the original information is still correctly recovered, and the effects can be reduced with proper system design.

Is there a time delay between the message and the demodulated signal? If so, what do you expect the reason behind it?

Yes, there is a time delay between the message signal and the demodulated signal. This delay is mainly caused by signal propagation time through the transmission medium and processing delay in the transmitter and receiver, such as filtering and demodulation circuits.

Is the time delay for the soft keying greater or smaller than that of the hard keying? Interpret your answer?

The time delay in soft keying is smaller than in hard keying because soft keying changes the signal amplitude gradually, which reduces abrupt transitions and transient effects. This smooth envelope allows the receiver's filters and envelope detector to settle more quickly, resulting in a shorter effective delay. In contrast, hard keying introduces sudden amplitude changes that create transients and ringing, increasing the settling time and thus causing a larger time delay.

Does the modulation index affect the time delay between the message and the demodulated signal?

The modulation index can influence the time delay between the message and demodulated signal, with higher indices potentially leading to longer delays due to increased complexity in modulation and demodulation processes.

Studying the effect of changing the message signal frequency on the demodulation:

The circuit connections were kept the same as in the previous parts of the experiment. The function generator was configured to produce a square-wave message signal with a frequency of 500 Hz, a peak-to-peak voltage of 10 V, and a duty cycle of 50%. The bipolar square-wave message signal was applied to the transmitter input (TXIN), and the Cassy sensor UA1 was connected at this point to observe the message signal. Another Cassy sensor, UB1, was connected to the receiver output (RXOUT) to measure and observe the demodulated signal.

(A_11)

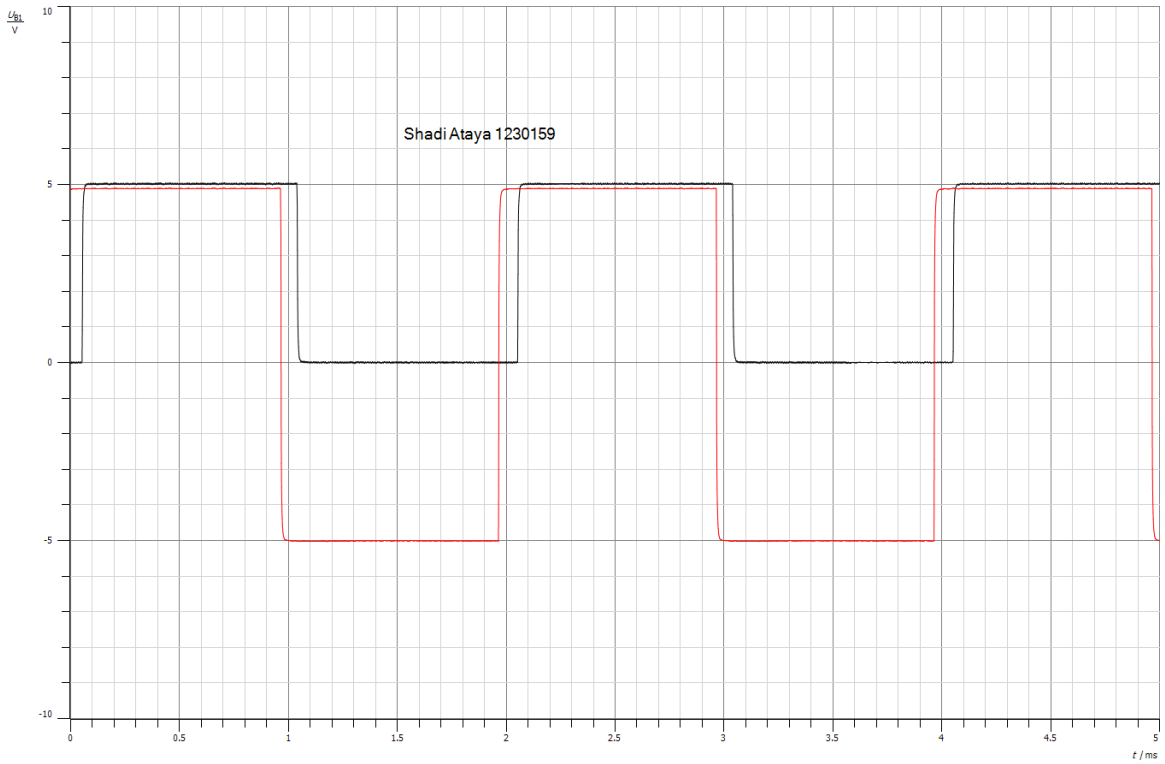


Figure 44: ASK demodulation with frequency = 500 Hz (A_11) in time domain

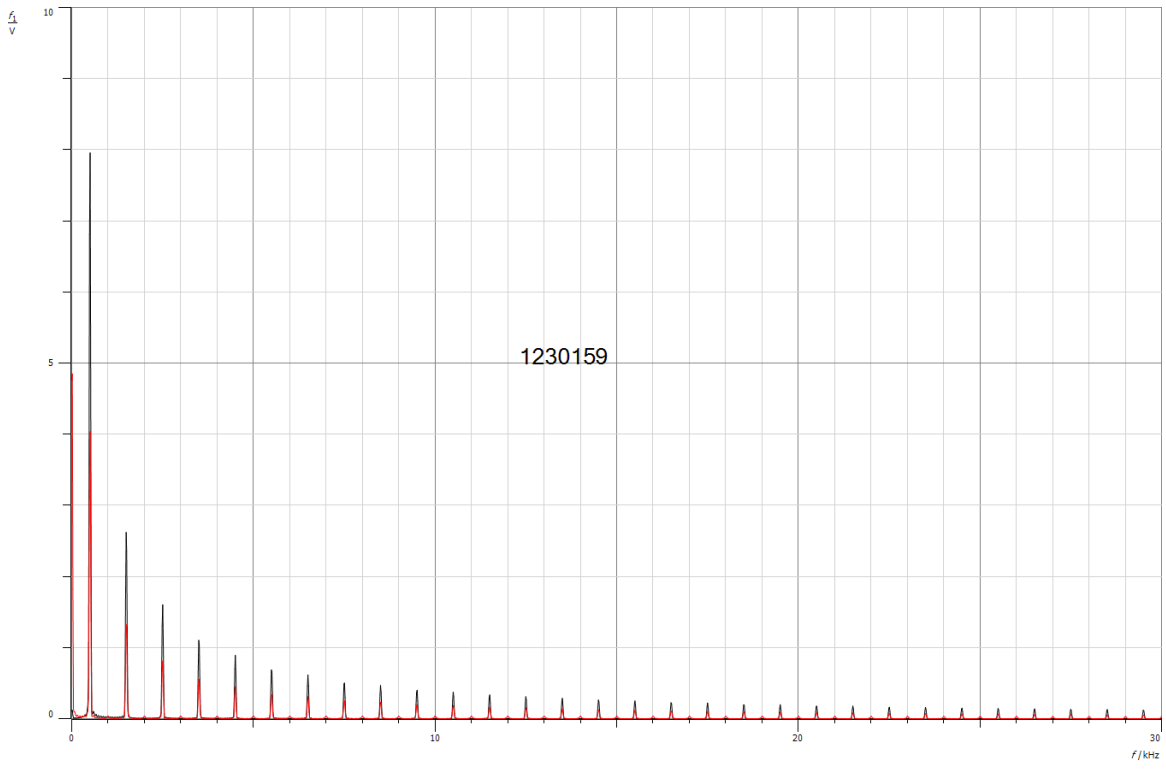


Figure 45: ASK demodulation with frequency = 500 Hz (A_11) in frequency domain

(A_21)

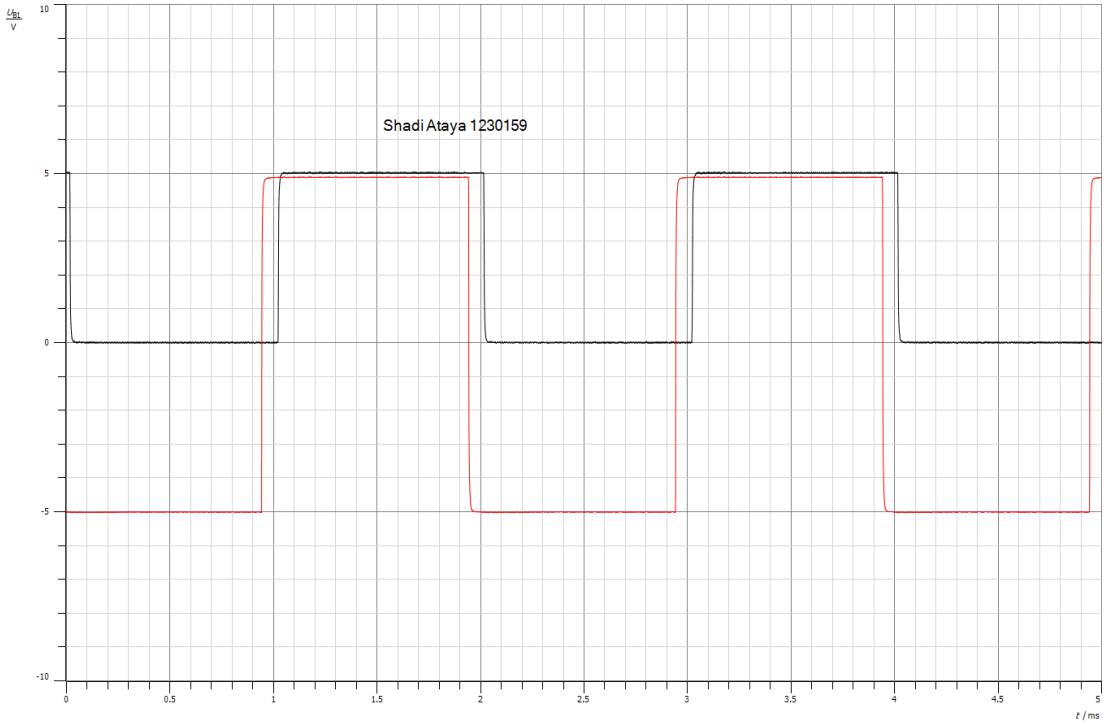


Figure 46: ASK demodulation with frequency = 500 Hz (A_21) in time domain

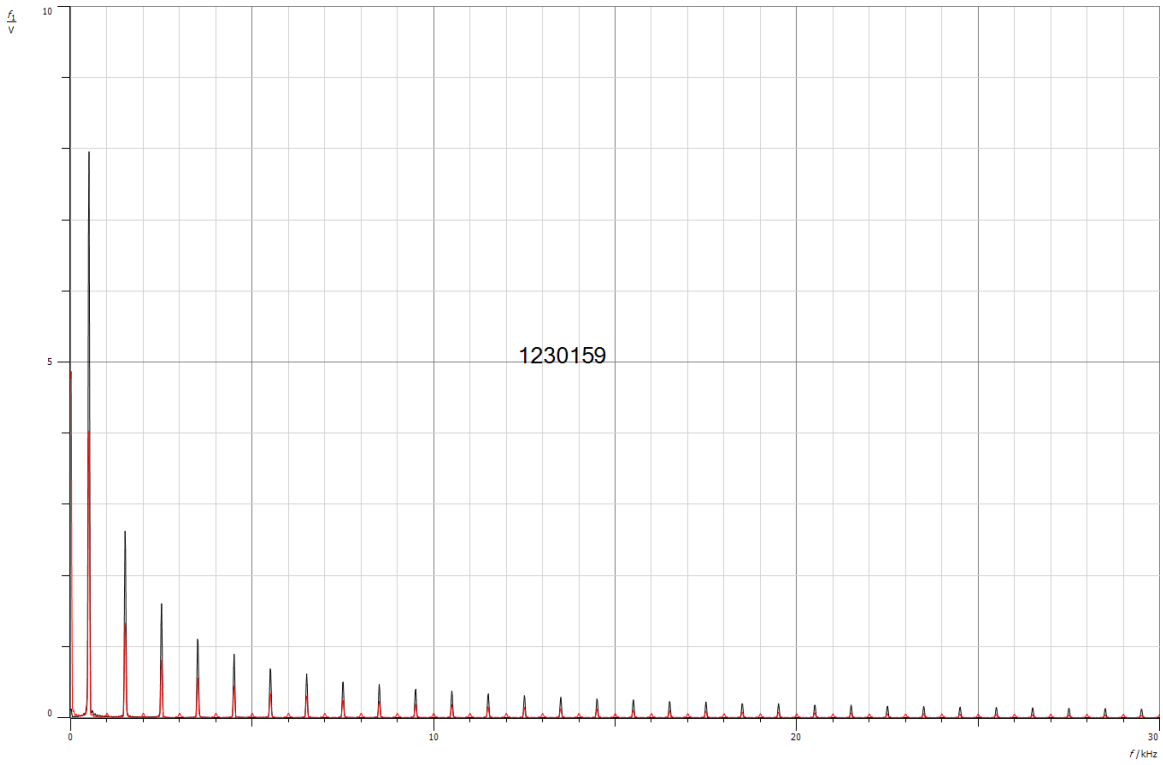


Figure 47: ASK demodulation with frequency = 500 Hz (A_21) in frequency domain

(A_12)

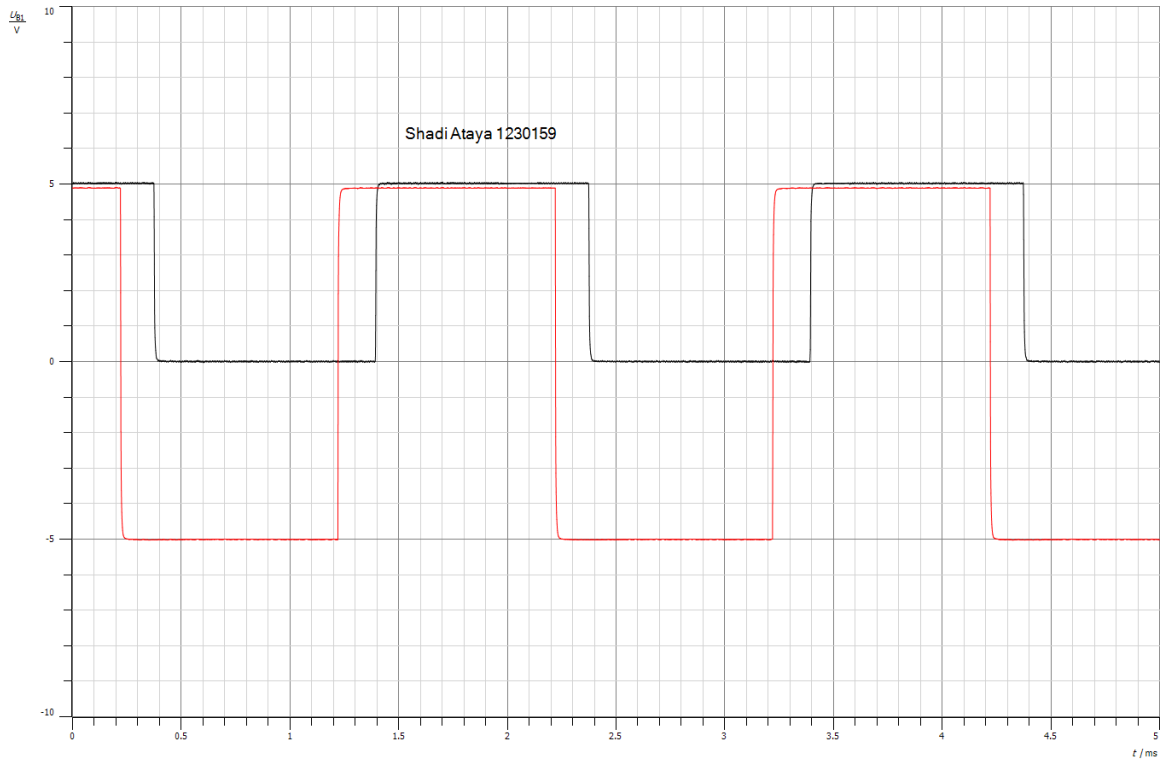


Figure 48: ASK demodulation with frequency = 500 Hz (A_12) in time domain

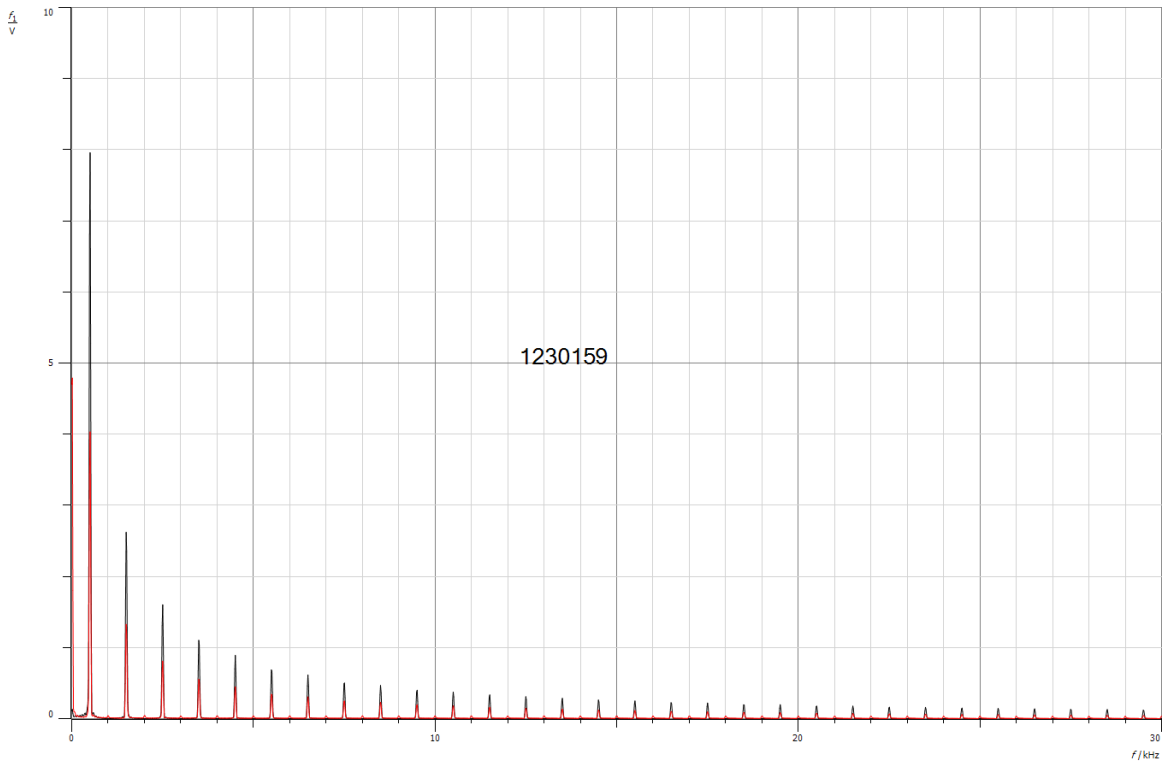


Figure 49: ASK demodulation with frequency = 500 Hz (A_12) in frequency domain

(A_22)

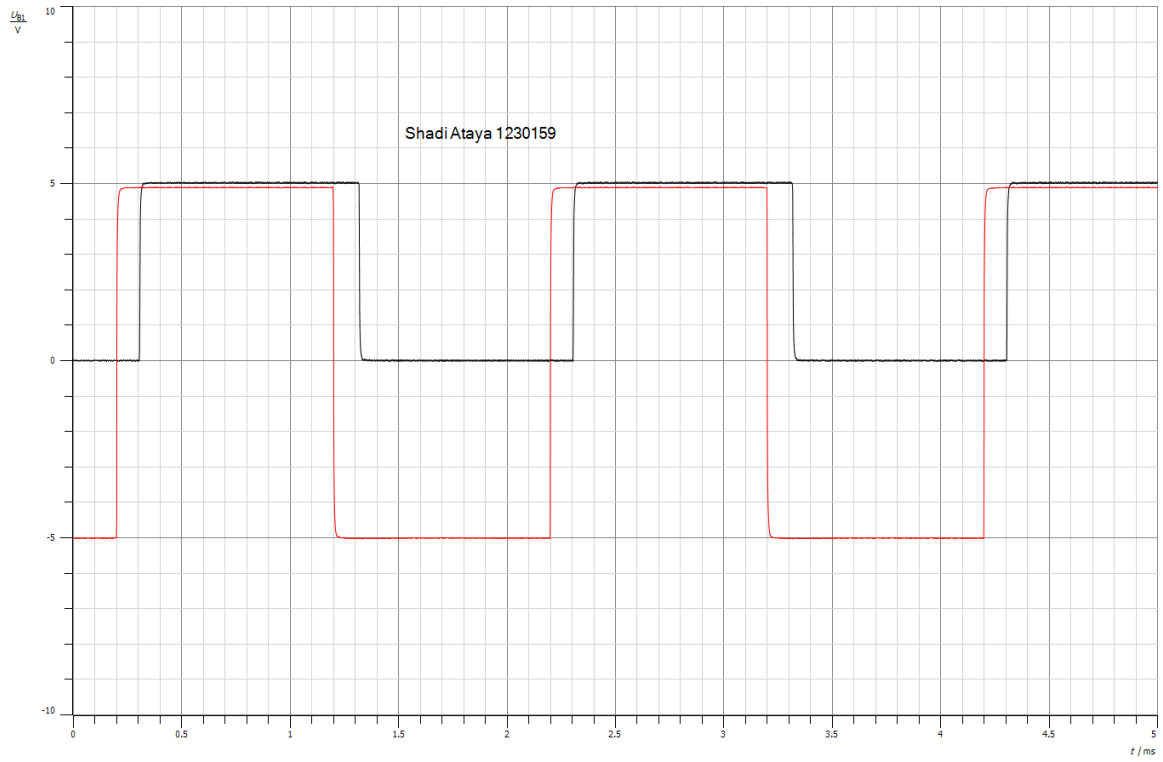


Figure 50: ASK demodulation with frequency = 500 Hz (A_22) in time domain

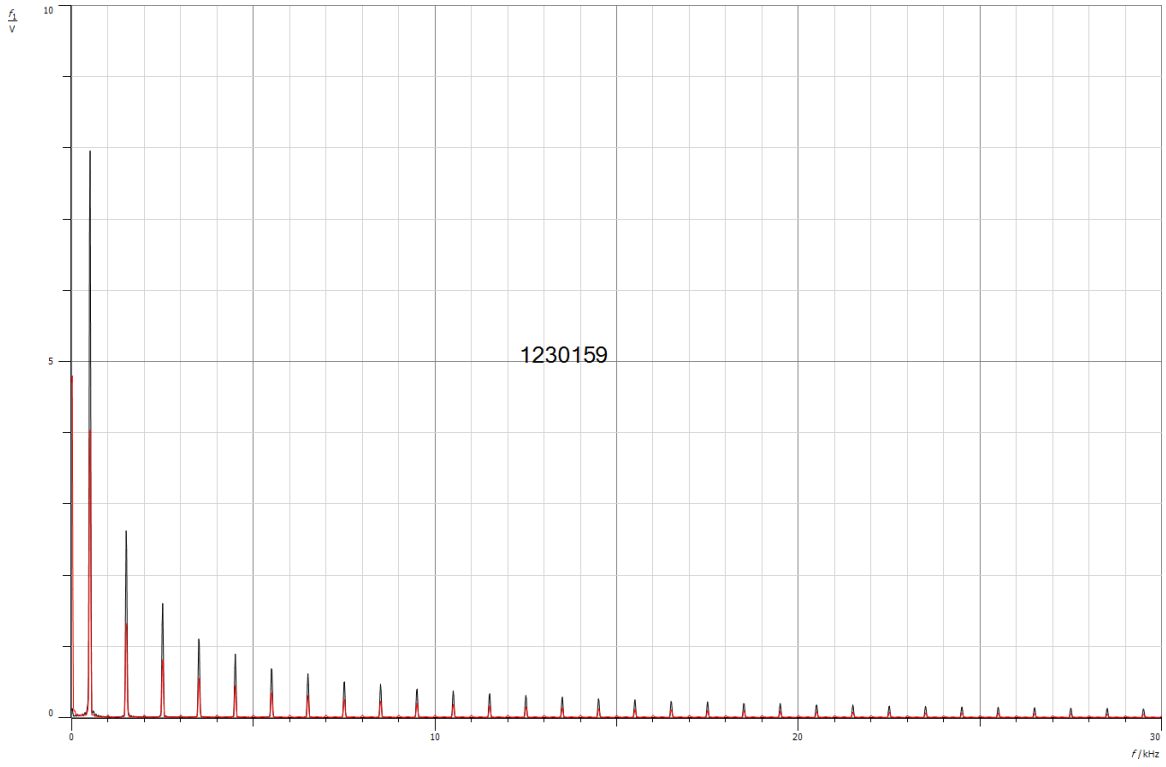


Figure 51: ASK demodulation with frequency = 500 Hz (A_22) in frequency domain

- Changing the message signal frequency in ASK directly affects demodulation performance. When the message frequency is increased, the demodulator must respond to faster envelope variations, which can introduce distortion, attenuation, and increased delay if the receiver bandwidth is limited. Lower message frequencies make demodulation easier and more accurate because the envelope changes more slowly, but they reduce the achievable data rate. Thus, there is a trade-off between demodulation quality and transmission speed when varying the message signal frequency.

Studying the effect of changing the message signal duty cycle on the demodulation:

The same connections as in the previous part were kept unchanged. The function generator was set to generate a square wave with a frequency of 1000 Hz, a peak-to-peak voltage of 10 V, and a duty cycle of 10%. The duty cycle was then gradually increased, and its effect on the ASK demodulation was observed.

(A_11)

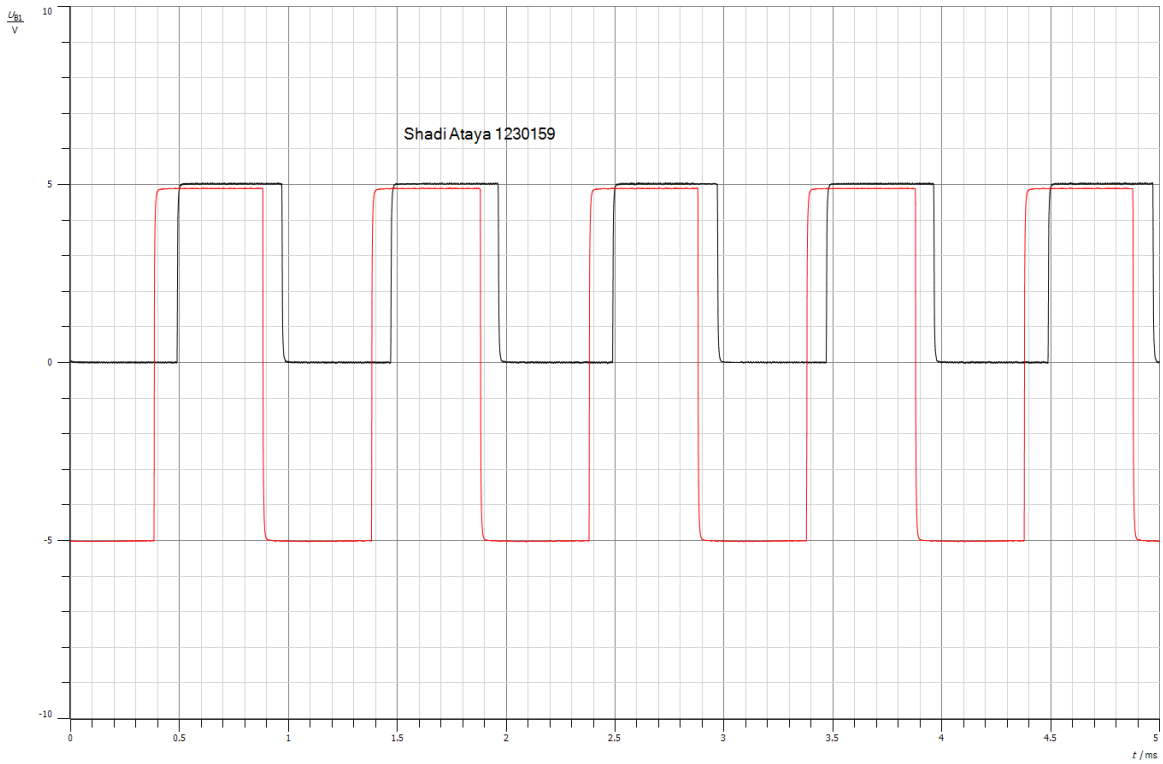


Figure 52: ASK demodulation with D = 10% (A_11) in time domain

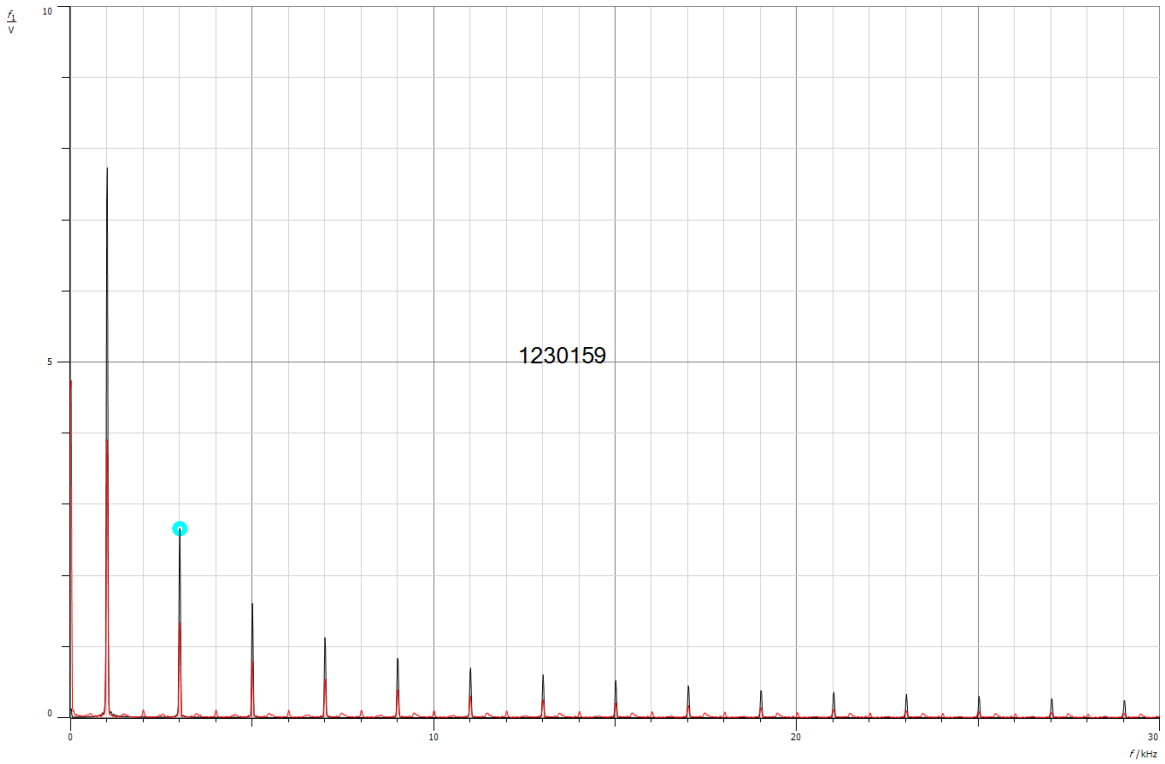


Figure 53: ASK demodulation with D = 10% (A_11) in frequency domain

(A_21)

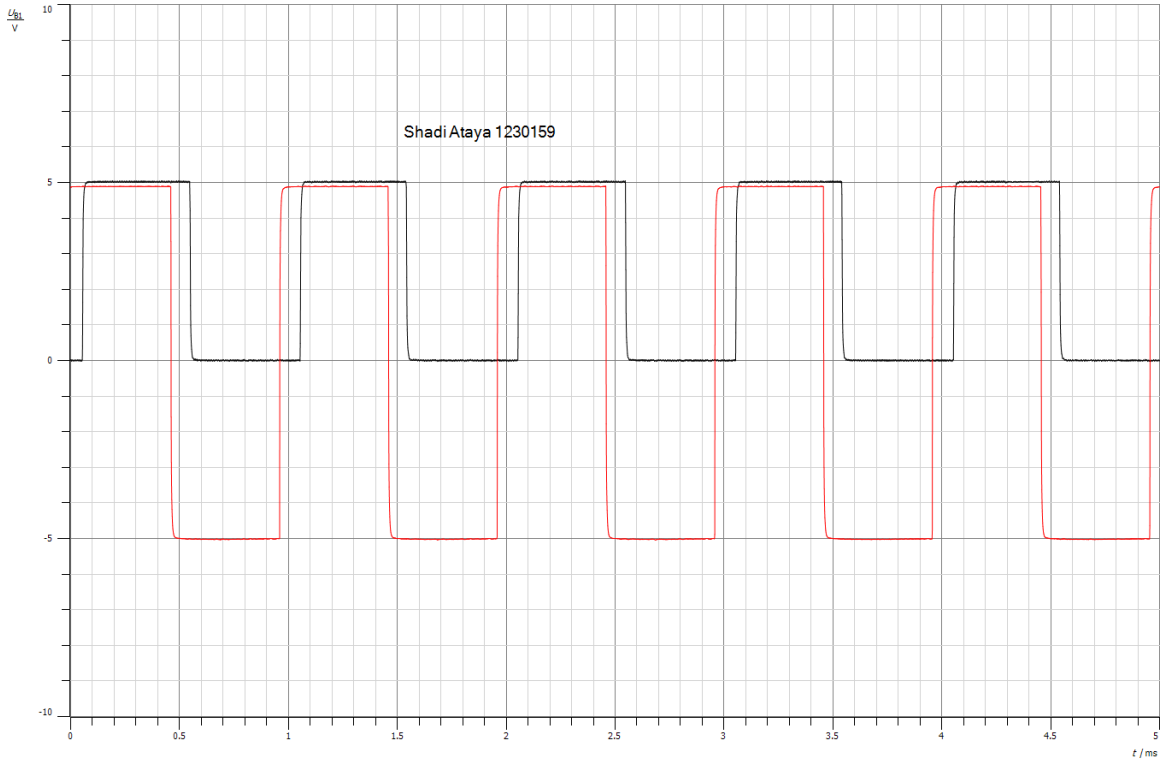


Figure 54: ASK demodulation with D = 10% (A_21) in time domain

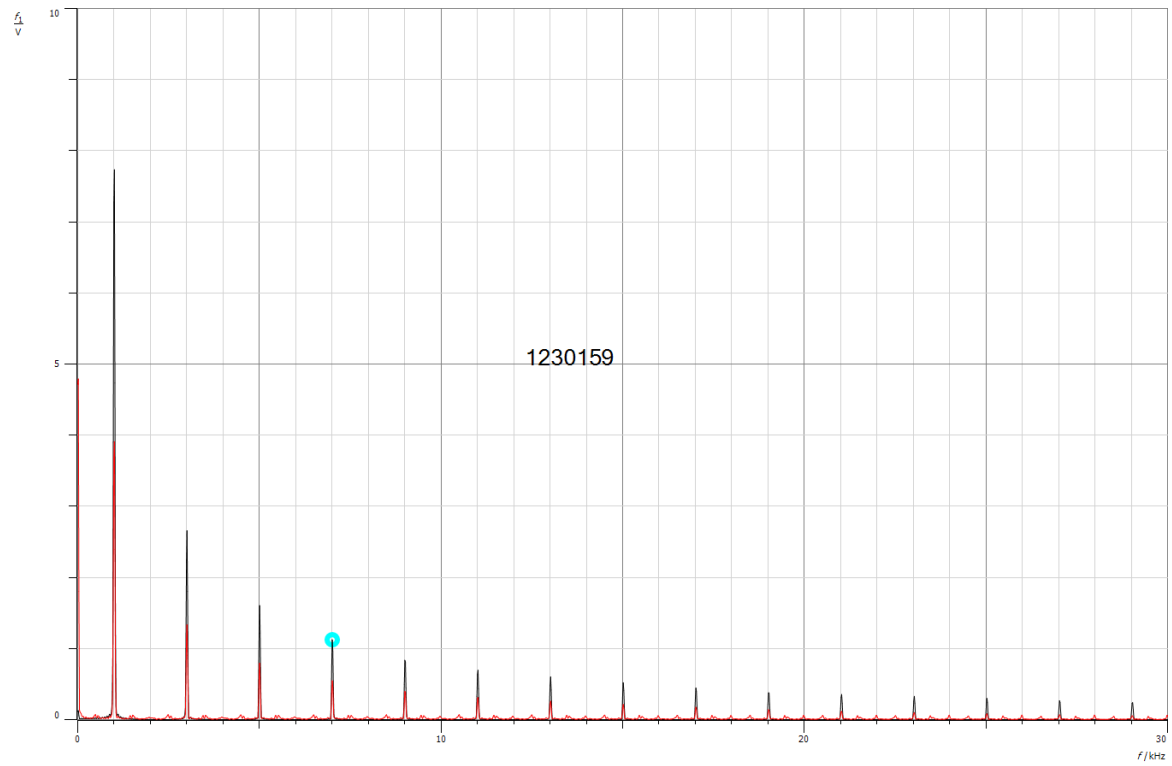


Figure 55: ASK demodulation with D = 10% (A_21) in frequency domain

(A_12)

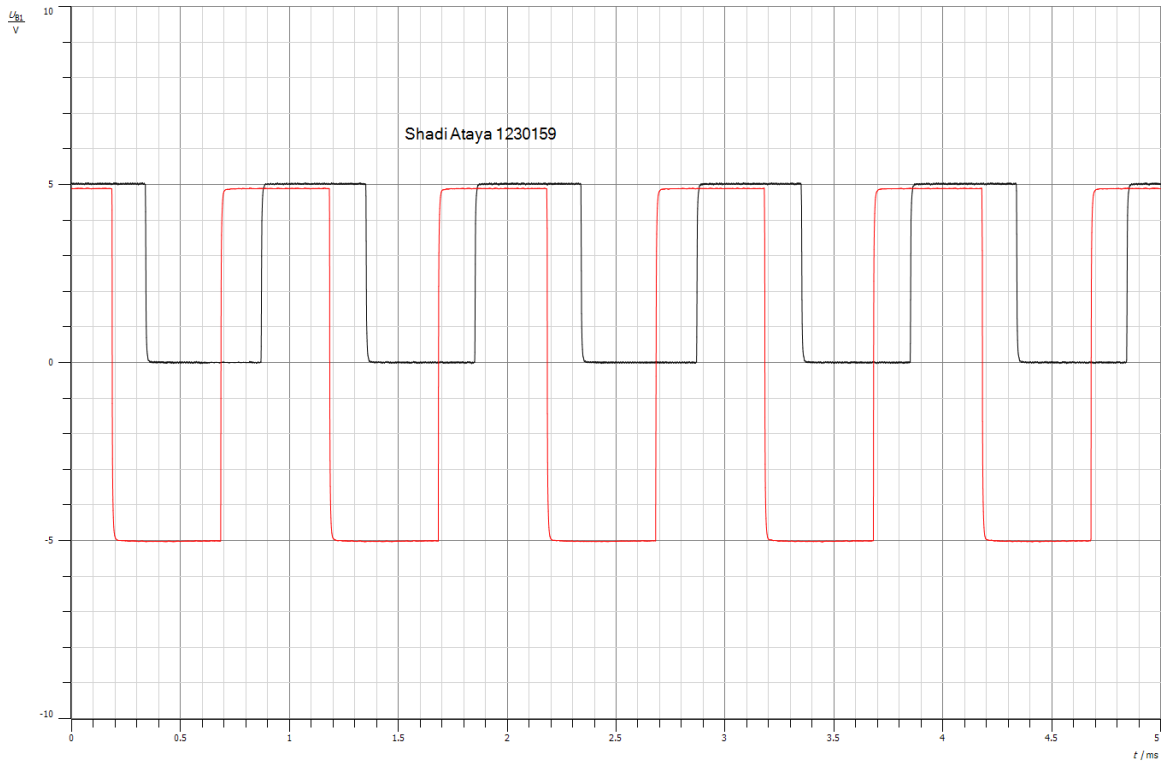


Figure 56: ASK demodulation with D = 10% (A_12) in time domain

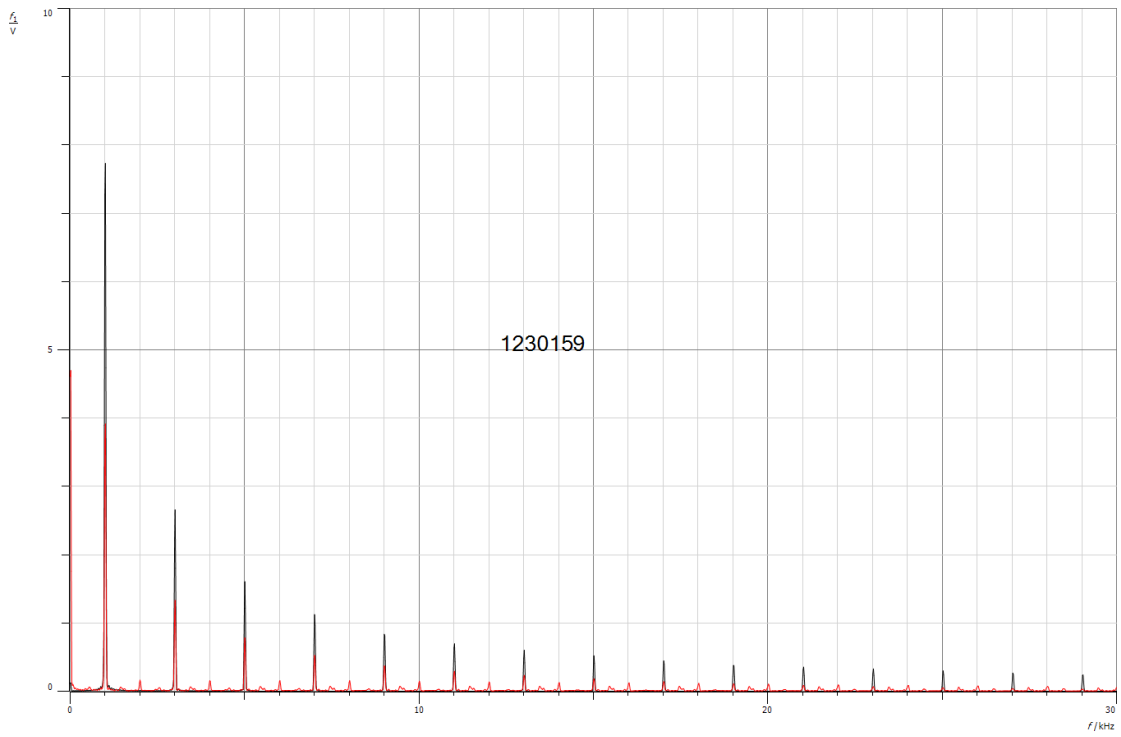


Figure 57: ASK demodulation with D = 10% (A_12) in frequency domain.

(A_22)

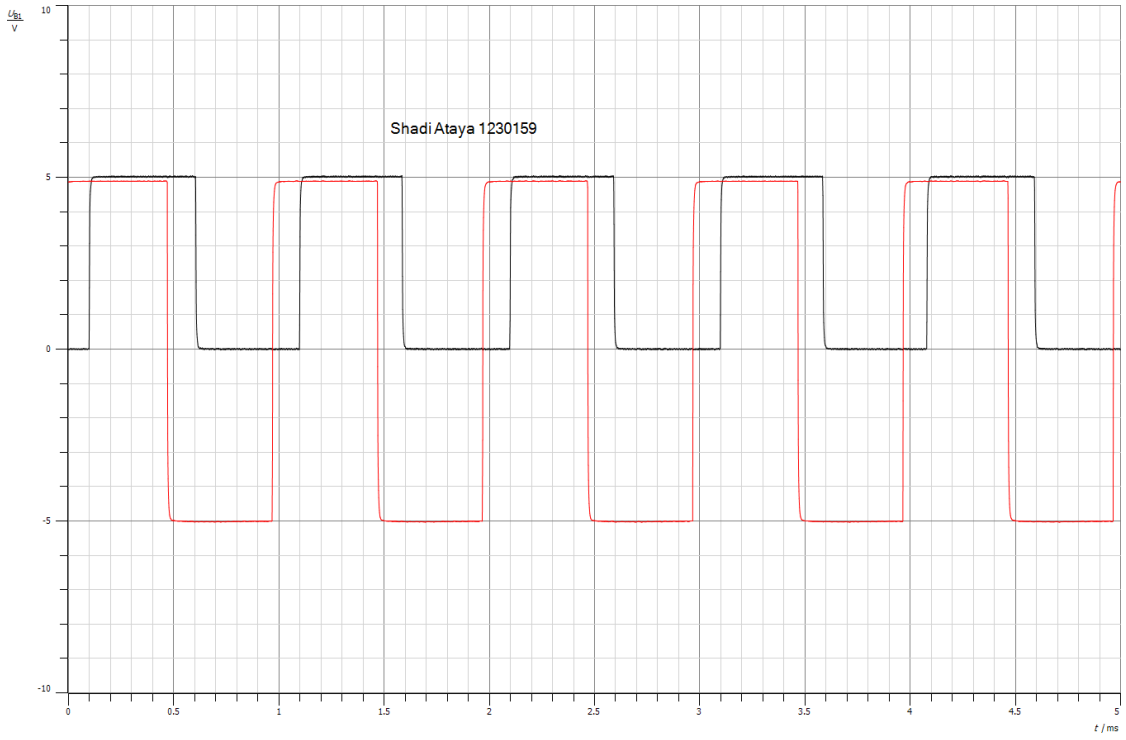


Figure 58: ASK demodulation with D = 10% (A_22) in time domain.

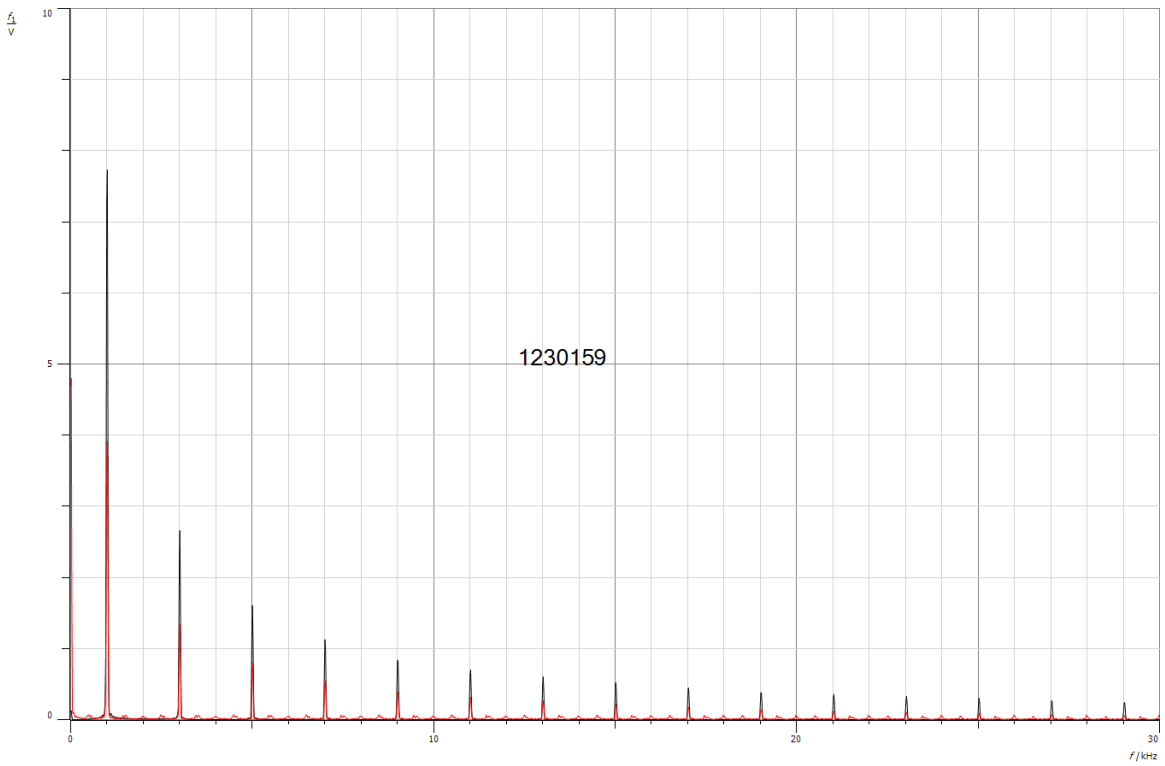


Figure 59: ASK demodulation with D = 10% (A_22) in frequency domain.

In ASK demodulation, the duty cycle of the message signal directly influences how clearly the envelope can be detected at the receiver. A high duty cycle improves demodulation because the carrier remains present for a longer time, producing a stronger and more stable envelope that is easier to track and less sensitive to noise. In contrast, a low duty cycle reduces demodulation performance since the carrier is present for a shorter duration, resulting in a weaker envelope, poorer tracking by the detector, and increased sensitivity to noise and distortion.

Questions

What duty-cycle value can be considered as the best one? Why?

The 50% duty cycle is considered the best for ASK demodulation because it provides a balanced on–off time for the carrier, producing a symmetric and stable envelope that the detector can track accurately. This balance minimizes distortion, reduces baseline variations, and offers good noise immunity, resulting in more reliable and accurate demodulation compared to very high or very low duty-cycle values.

Is there a difference between the smaller and greater duty-cycle values? Will they affect the low or high state of the Ask signal?

Yes, there is a difference between smaller and greater duty-cycle values in ASK. A greater duty cycle keeps the carrier on for a longer time, which strengthens the high (on) state of the ASK signal and makes it easier to detect at the receiver. A smaller duty cycle keeps the carrier on for a shorter duration, which weakens the high state and increases sensitivity to noise and distortion. The low (off) state is much less affected, since it represents little or no carrier presence, so duty-cycle variations mainly impact the quality and detectability of the high state of the ASK signal.

Conclusion

In conclusion, this experiment provided a comprehensive study of Amplitude Shift Keying (ASK) modulation and demodulation using both hard keying (A-11, A-21) and soft keying (A-12, A-22) with modulation indices of 100% and 50%. The differences between hard and soft keying were clearly demonstrated, showing that while soft keying is more complex to implement, it offers smoother signal transitions and behaviour closer to theoretical expectations. The effects of modulation index, threshold level, and message signal parameters were examined in both time and frequency domains, revealing their influence on bandwidth, spectral components, and signal detectability. Changes in message frequency and duty cycle were found to affect demodulation quality, time delay, and distortion, particularly at higher frequencies and extreme duty-cycle values. Overall, the practical results closely matched theoretical predictions, providing valuable insight into the performance trade-offs and parameter selection needed for efficient ASK communication systems.

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[3] https://www.fritz.dellsperger.net/downloads/V7a_3-Digital%20Modulation_en.pdf

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